Modulation Technique, Channel and Speech Coding

Chapter 4

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Outline

1. Modulation
2. Digital modulations
   - Linear Modulation
   - Non-Linear Modulation
3. Spread Spectrum Modulation Techniques
   - Frequency Hopped Spread Spectrum (FH-SS)
   - Direct Sequence Spread Spectrum (DS-SS)
4. Orthogonal Frequency Division Multiplexing (OFDM)
5. Speech Coding

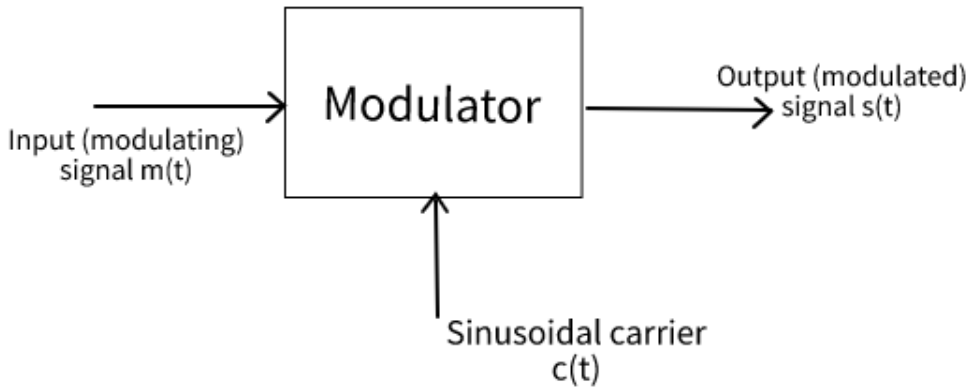
Modulation

- Modulation is the process of encoding information from a message source in a manner suitable for transmission
  - some characteristic of a carrier wave is varied in accordance with an **information-bearing signal**, i.e., baseband message signal
- It involves translating a **baseband message signal** to a **bandpass signal** at frequencies that are very high compared to the baseband frequency
- General sinusoid $A_c \cos(2\pi f_c t + \phi(t))$
  - A_c : Amplitude
  - f_c : Carrier frequency
  - ϕ : Phase
  - t : time

Benefits from Modulation

1. It is used to shift the spectral content of a message signal so that it lies inside the operating frequency band of the wireless communication channel
   - Ex.: telephonic communication over cellular radio channel
     Voice $\approx 300 - 3100$ Hz $\rightarrow$ freq. assigned to cellular radio channel $\approx 900 - 1800$ MHz
2. It provides a mechanism for putting the information content of a message signal into a form that be less vulnerable to noise or interference
   - Received signal ordinarily corrupted by noise $\rightarrow$ Modulation : improve system performance in presence of noise
3. It permits the use of multiple-access techniques
   - Simultaneous transmission of several different information-bearing signals over the same channel

Principal characteristics



Linear Modulation :

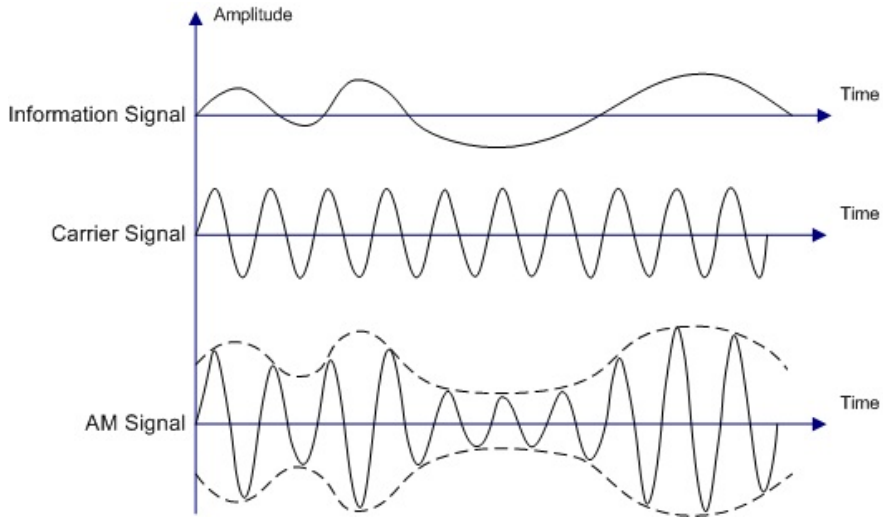
- Input-Output relation of modulator satisfies **principle of superposition**
  - Output of modulator produced by a number of inputs applied simultaneously is equal to the sum of the output that result when the inputs are applied one at a time
    $M(i_1 + \cdots + i_n) = M(i_1) + \cdots + M(i_n)$

Nonlinear Modulation:

- Input-Output relation of modulator does not satisfies **principle of superposition**

Type of Modulation

1. Analog Modulation
   - If the variation in the parameter of the carrier is continuous (input analog signal)



2. Digital Modulation
   - If the variation in the parameter of the carrier is discrete (input discrete/digital signal)

Amplitude Modulation (AM)

- The process in which the amplitude of a carrier signal is varied in accordance with the message signal is called the amplitude modulation.
- If $A_c \cos(2\pi f_c t)$ is the carrier signal and $m(t)$ is the message signal, then the AM signal can be represented as

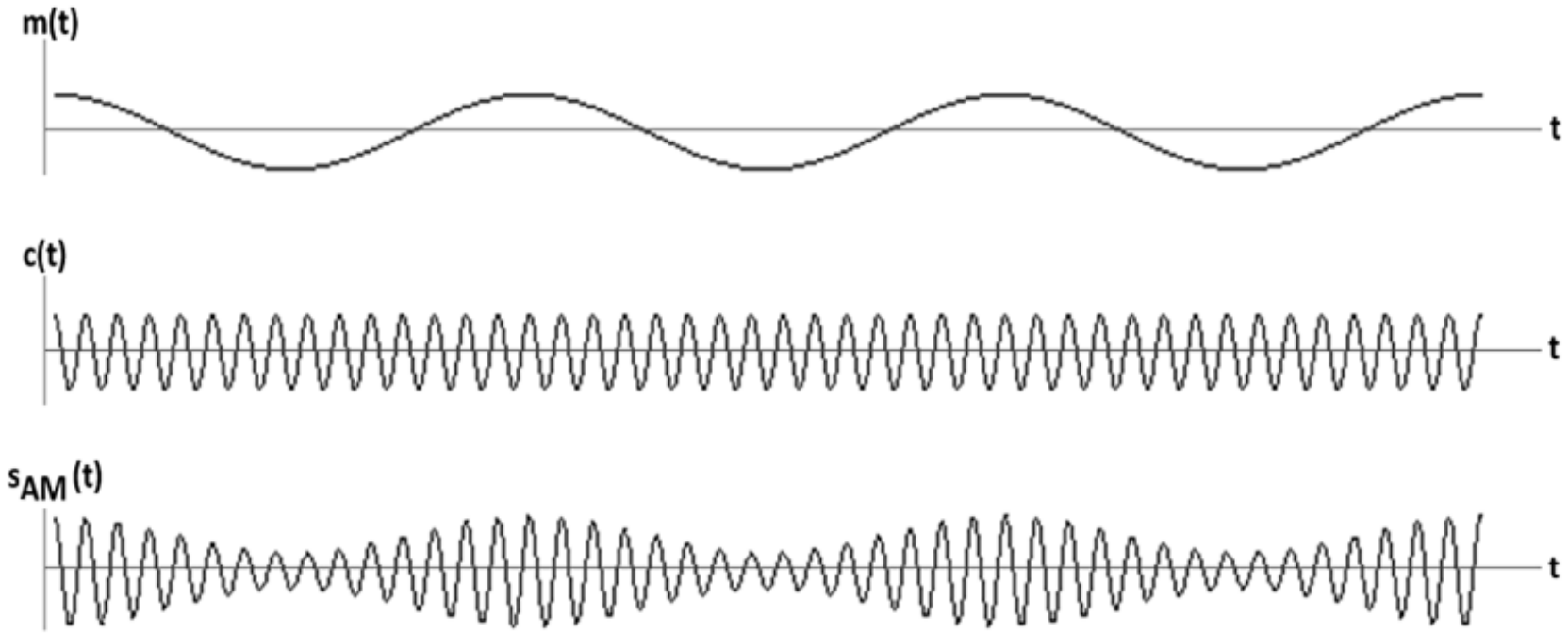
  $S_{AM}(t) = A_c[1 + m(t)] \cos(2\pi f_c t)$
- The spectrum of an AM signal can be shown to be

  $S_{AM}(f) = \frac{A_c}{2}[\delta(f - f_c) + M(f - f_c) + \delta(f + f_c) + M(f + f_c)]$
- The bandwidth of an AM signal is twice the bandwidth of the message signal.
- The total power in an AM signal can be shown to be

  $P = P_c(1 + \mu^2/2)$

  $P_c = A_c^2/2$ is the carrier power

Amplitude Modulation



Digital Modulations

- In digital Modulation: an analog carrier signal is modulated by a discrete signal
  - Digital modulation: digital to analog
  - Demodulation: analog to digital
- The modulating wave consists of binary data and the carrier is sinusoidal wave
- If the information is digital changing parameters is called "keying"
- The modulating signal represents as time sequence of symbols or pulses
  - Each symbol has k finite states
  - Each symbol represents n bits of information where $n = \log_2 k$ bits/symbol
- The source information is normally represented as a baseband (low-pass) signal $m(t)$

Linear Modulation

- The amplitude of the transmitted signal $s(t)$, varies linearly with the modulating signal $m(t)$
- bandwidth efficient
- Examples:
  - Binary Phase Shift Keying (BPSK)
  - Differential Phase Shift Keying (DPSK)
  - Quadrature Phase Shift Keying (QPSK)

Binary Phase Shift Keying (BPSK)

- Binary data are represented by two signals with different phases
- Phase ϕ of carrier is changed in response to the information
  - One phase change encodes 0 while another phase change encodes 1
- Carrier signal: $A_c \cos(2\pi f_c t + \phi(t))$
- The modulated signal is

$$s(t) = \begin{cases} A_c \cos(\omega_c t + 0) = A_c \cos(2\pi f_c t) & \text{symbol "1"} \\ A_c \cos(\omega_c t + \pi) = -A_c \cos(2\pi f_c t) & \text{symbol "0"} \end{cases}$$

- These two signals have the same frequency and energy
  - Two signals have a correlation coefficient of -1, which leads to the minimum error probability for the same $\frac{E_b}{N_0}$

Binary Phase Shift Keying (BPSK)

Let E_b is energy per bit, T_b is bit period

$$E_b = \int_0^{T_b} s^2(t)dt = \int_0^{T_b} A_c^2 \cos^2(2\pi f_c t)dt$$

$$= A_c^2 \int_0^{T_b} \frac{1+\cos(4\pi f_c t)}{2} dt = \frac{A_c^2}{2} T_b \quad [\because \int_0^{T_b} \cos(4\pi f_c t)dt = 0]$$

$$A_c = \sqrt{\frac{2E_b}{T_b}}$$

$$s(t) = \begin{cases} \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t) & \text{symbol "1"} \\ -\sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t) & \text{symbol "0"} \end{cases}$$

$$s(t) = \{\sqrt{E_b}, -\sqrt{E_b}\}$$

$$\phi(t) = \sqrt{\frac{2}{T_b}} \cos(2\pi f_c t)$$

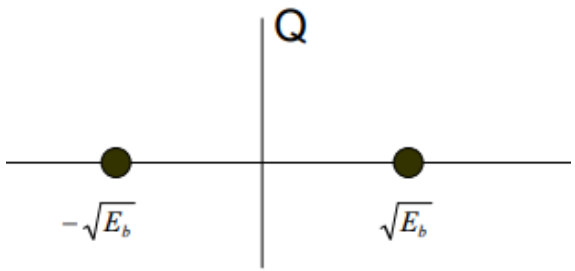
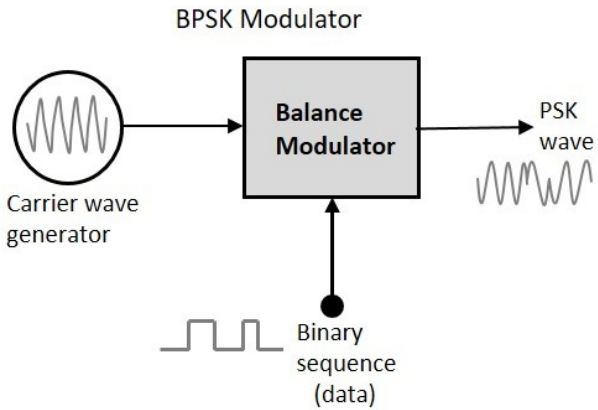
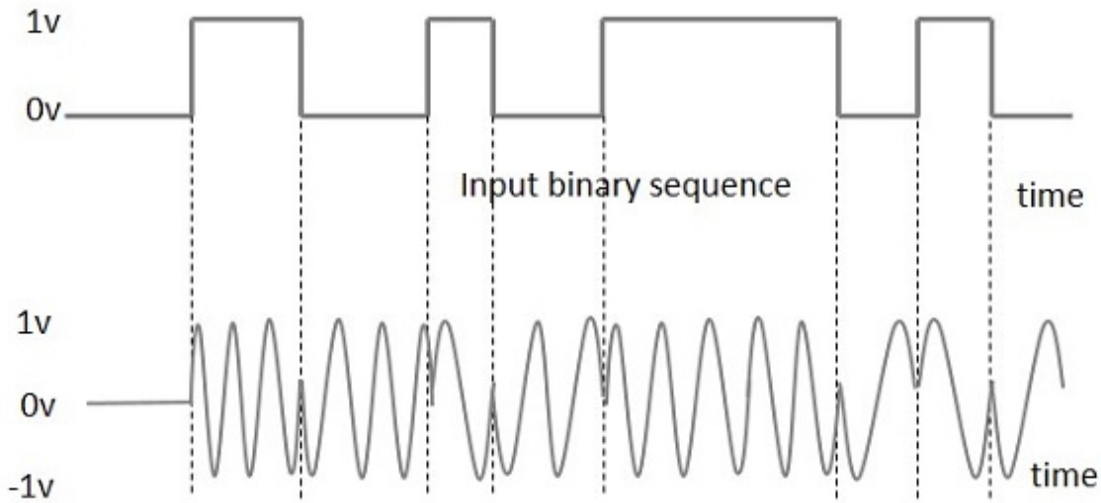


Figure: Constellation

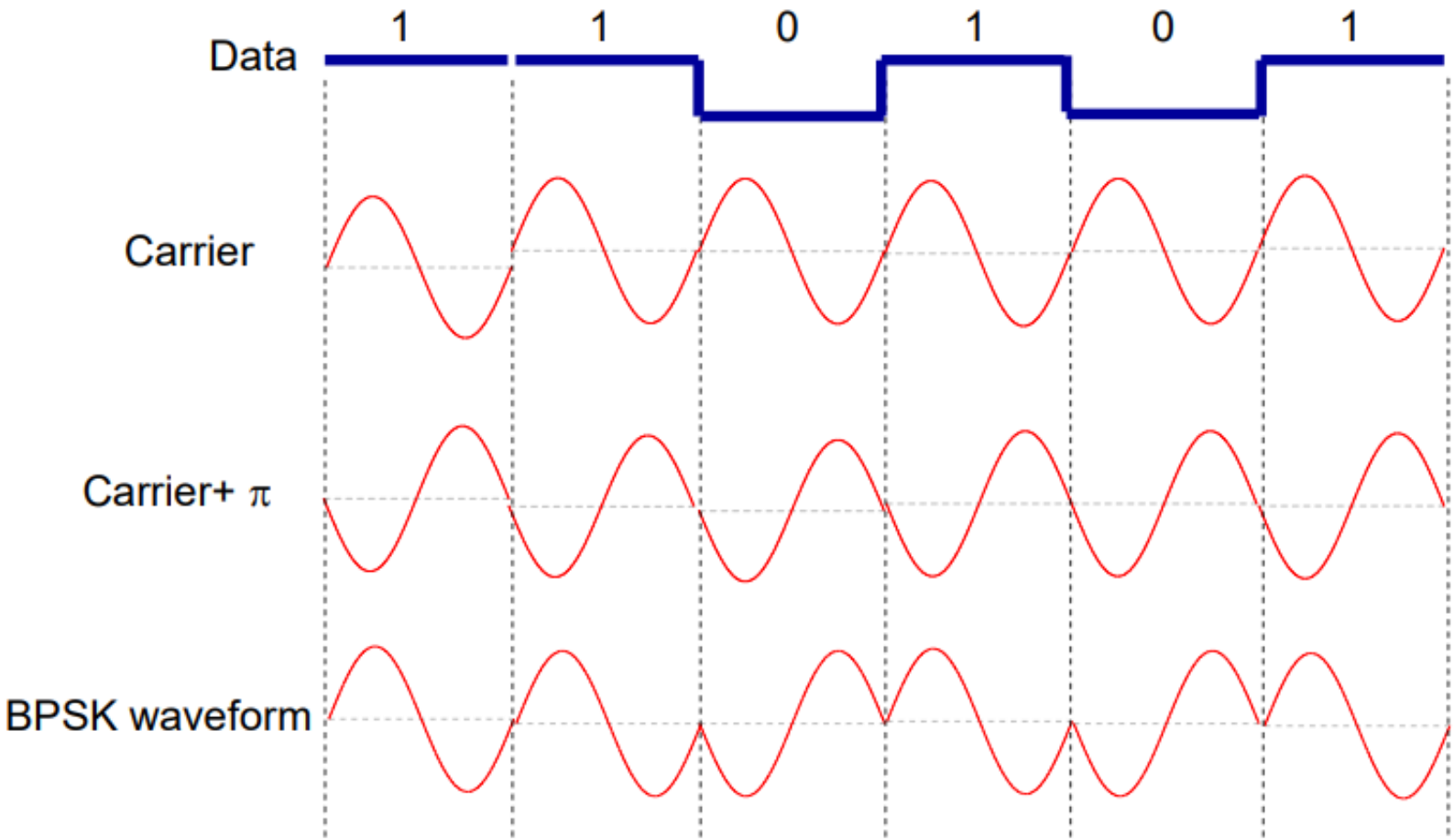
BPSK transmitter



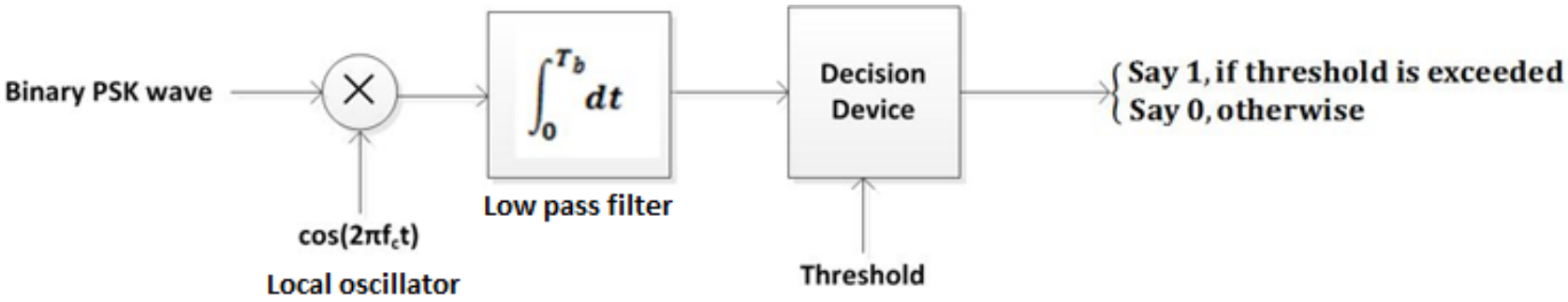


BPSK Modulated output wave

Example



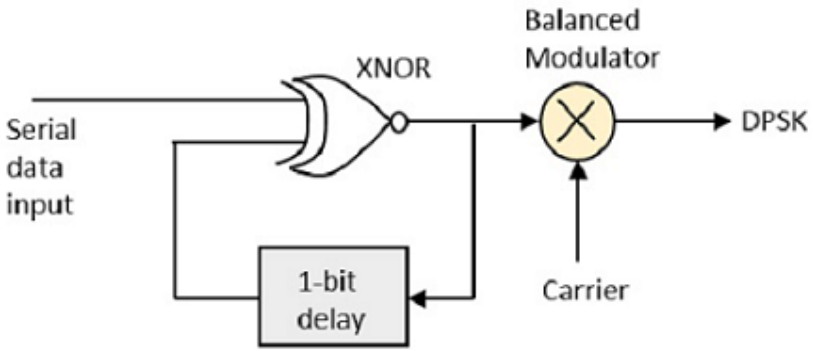
BPSK Receiver



- The baseband signal ($m(t)$): uniquely recovered from $s(t)$ by multiplying locally generated oscillator and then low-pass filtering the product
- It is assumed that the local oscillator signal is exactly coherent or synchronized, in both frequency and phase with the carrier wave $c(t)$
  - This method of demodulation is known as coherent demodulation

Differential Phase Shift Keying (DPSK)

- DPSK is a technique of BPSK, in which there is no reference phase signal i.e., non-coherent demodulator
- The transmitted signal itself can be used as a reference signal
- DPSK encodes two distinct signals, i.e., the carrier and the modulating signal with 180° phase shift each
- The serial data input is given to the XNOR gate and the output is again fed back to the other input through 1-bit delay
- The output of the XNOR gate along with the carrier signal is given to the balance modulator, to produce the DPSK modulated signal



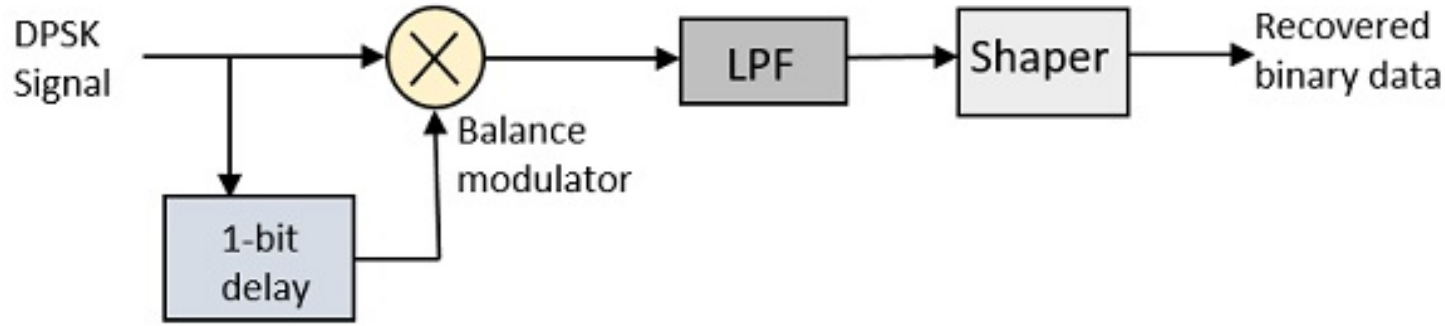
DPSK Modulator

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

DPSK Code

Binary Data		0	0	1	0	0	1	0	0	1	1
Differentially Encoded Data	1	0	1	1	0	1	1	0	1	1	1
Phase of DPSK	0	π	0	0	π	0	0	π	0	0	0
Shifted Differentially encoded Data d_{k-1}		1	0	1	1	0	1	1	0	1	1
Phase of shifted Data		0	π	0	0	π	0	0	π	0	0
Phase Comparision Output		-	-	+	-	-	+	-	-	+	+
Detected Binary Seq.		0	0	1	0	0	1	0	0	1	1

DPSK Demodulator



- The received DPSK signal is applied to one input of the multiplier and a delayed version of the received DPSK signal by the time interval T_b is applied to the other input of the multiplier
- The output of the difference is proportional to $\cos(\phi)$
  - ϕ is the difference between the carrier phase angle of the received DPSK signal and its delayed version, measured in the same bit interval

DPSK Demodulator

- The phase difference $\Delta\phi$ between the two sequences for each bit interval is used to determine the sign of the phase comparator output.
  - When $\Delta\phi = 0$, the integrator output is positive
  - When $\Delta\phi = \pi$ the integrator output is negative
- By comparing the integrator output with a decision level of zero volt
  - decision device can reconstruct the binary sequence by assigning
    - a symbol '0' for negative output
    - a symbol '1' for positive output

DPSK Advantages and Disadvantages

Advantages

- Demodulation doesn't need the carrier signals at the end of the receiver circuit
  - receivers are simple and inexpensive to construct
- Bandwidth of DPSK requirement is low evaluated to BPSK modulation

Disadvantages

- Bit error rate or chance of error is high in DPSK contrast to BPSK
  - error in primary bit makes error within a subsequent bit as well as consecutively error spreads

Quadrature Phase Shift Keying (QPSK)

- Variation of BPSK
  - Multilevel Modulation Technique: 2 bits per symbol
- **Two bits per symbol** with a minimum phase separation of $\frac{\pi}{2}$
  - Modulating signals onto carrier signal using four phase 0, $\frac{\pi}{2}$, π, $\frac{3\pi}{2}$ states to code two digital bits
- The modulated signal is $S_i(t) = A_c \cos(2\pi f_c t + \phi_i(t))$, $1 \leq i \leq 4$

Dibit	Phase $\phi(t)$
00	0
01	$\frac{\pi}{2}$
10	π
11	$\frac{3\pi}{2}$

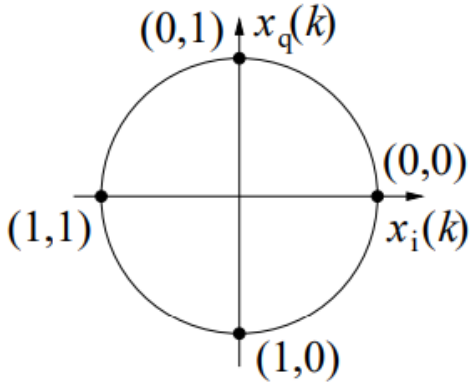
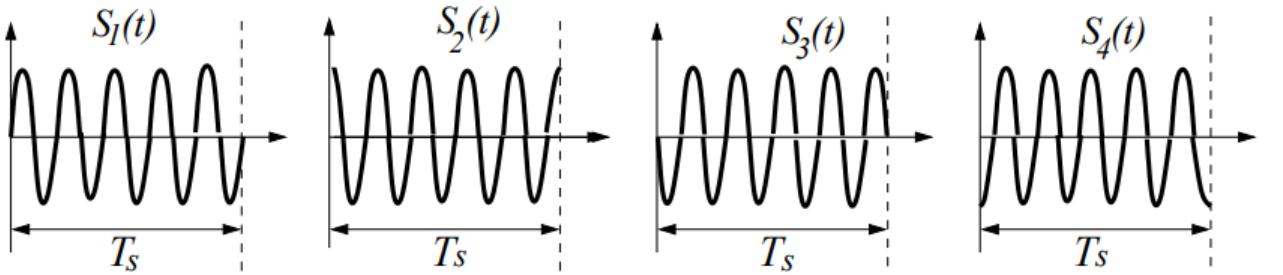
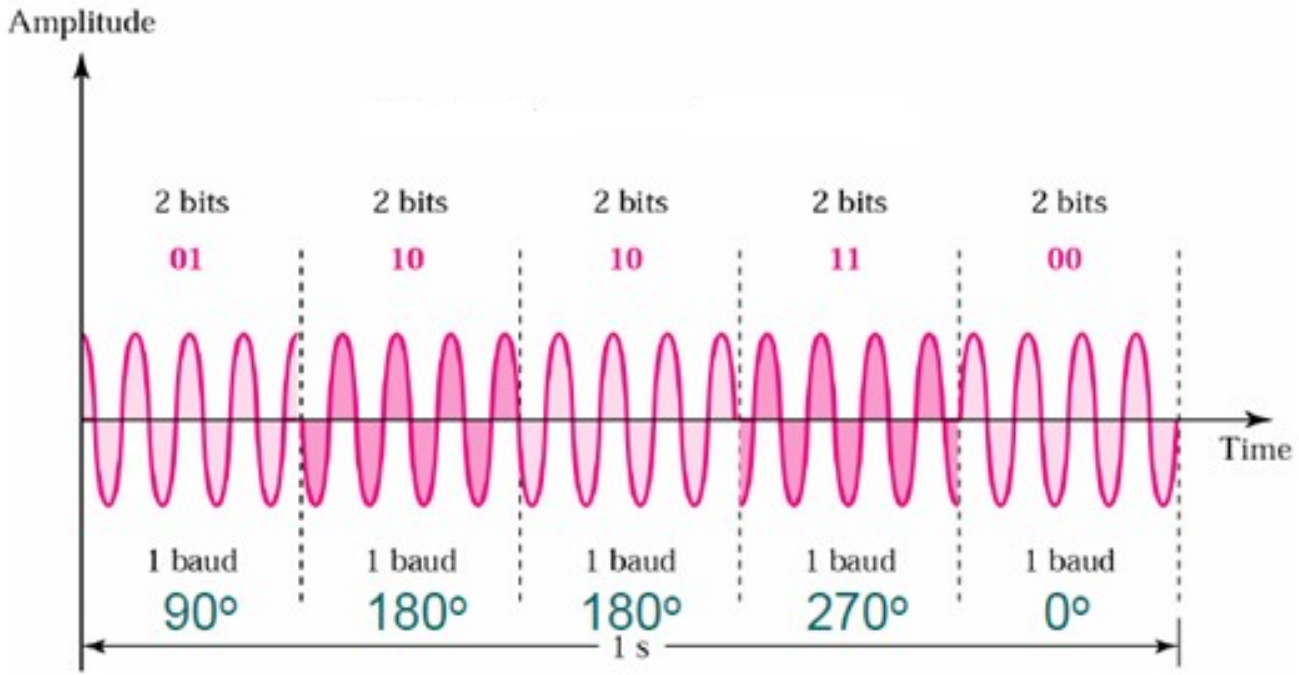


Figure: Constellation

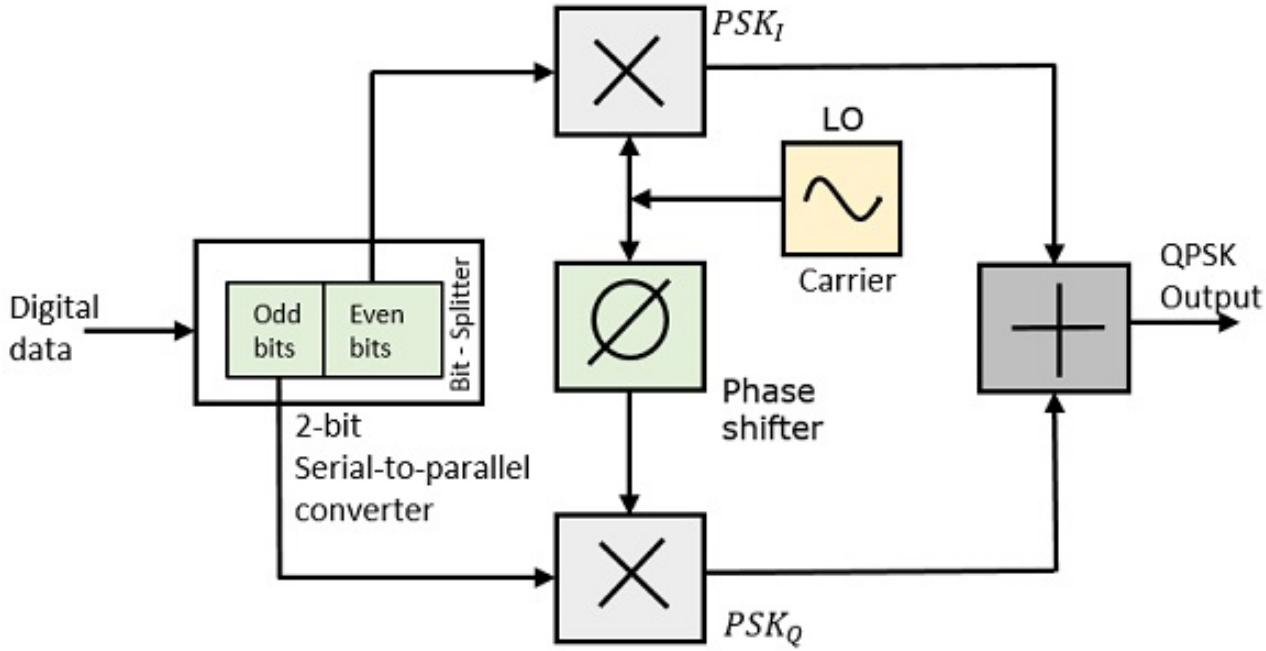
Quadrature Phase Shift Keying (QPSK)



bit (0,0): $\phi_1 = 0$ bit (0,1): $\phi_2 = \frac{\pi}{2}$ bit (1,1): $\phi_3 = \pi$ bit (1,0): $\phi_4 = \frac{3\pi}{2}$



QPSK Modulator

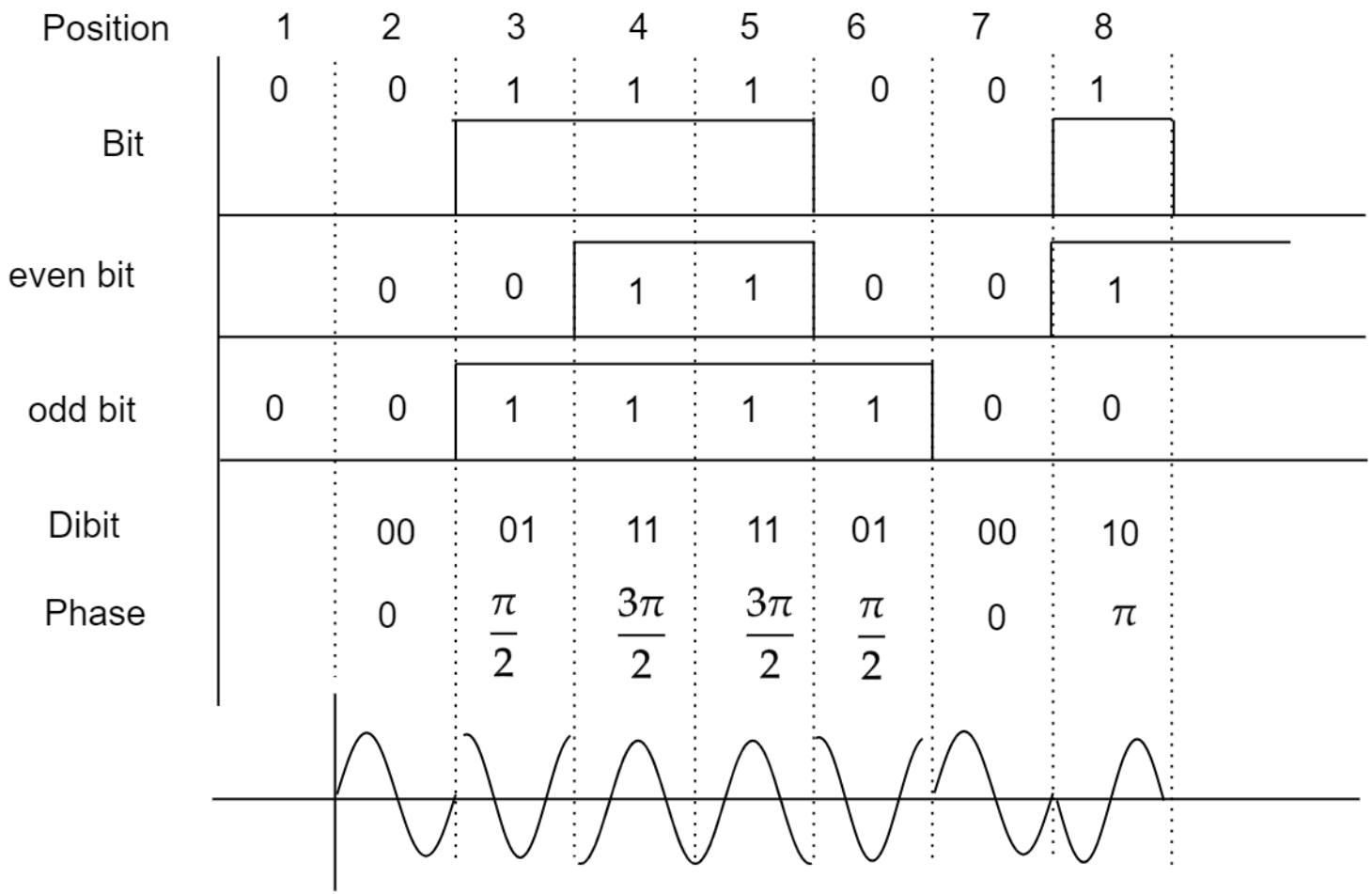


The QPSK Modulator uses a bit-splitter, two multipliers with local oscillator, a 2-bit serial to parallel converter, and a summer circuit

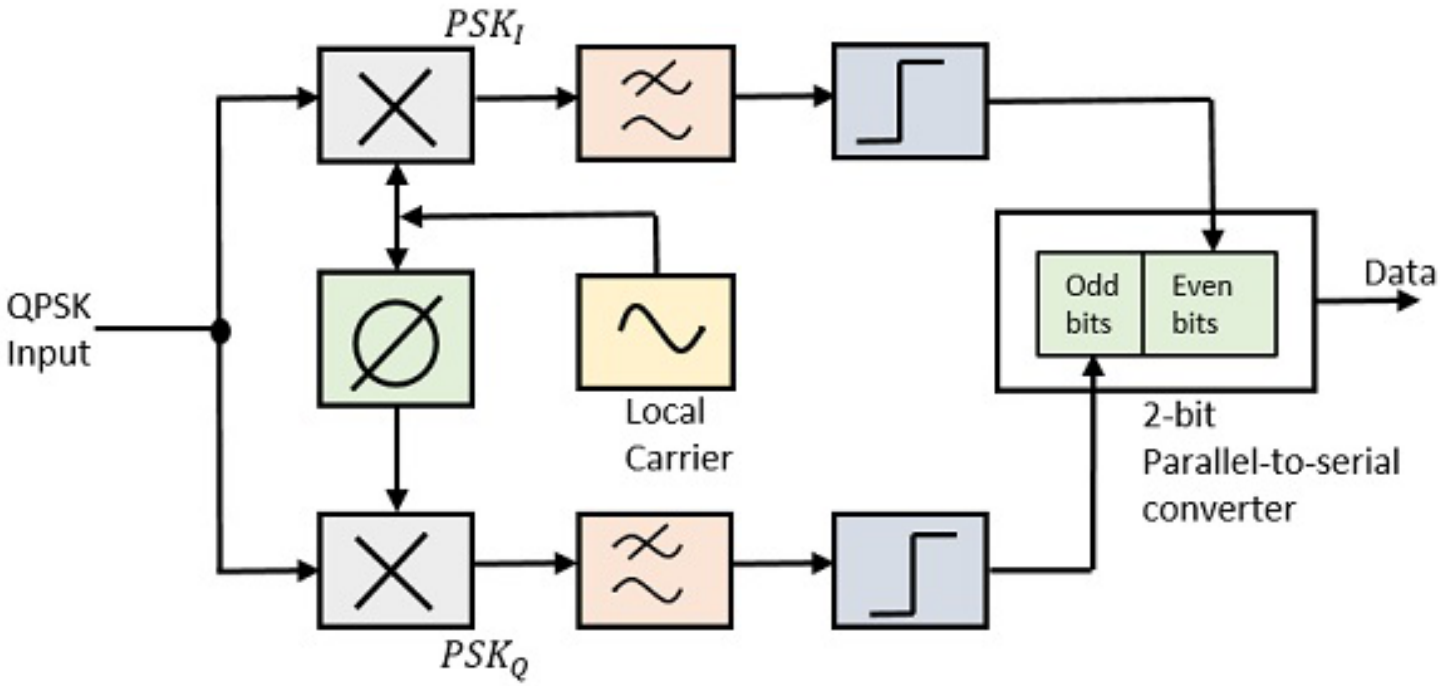
QPSK Modulator

- At the modulator's input, the message signal's even bits (i.e., 2nd bit, 4th bit, 6th bit, etc.) and odd bits (i.e., 1st bit, 3rd bit, 5th bit, etc.) are separated by the bits splitter
  - are multiplied with the same carrier to generate odd BPSK (called as PSKI) and even BPSK (called as PSKQ)
- The PSKQ signal is anyhow phase shifted by 90° before being modulated.

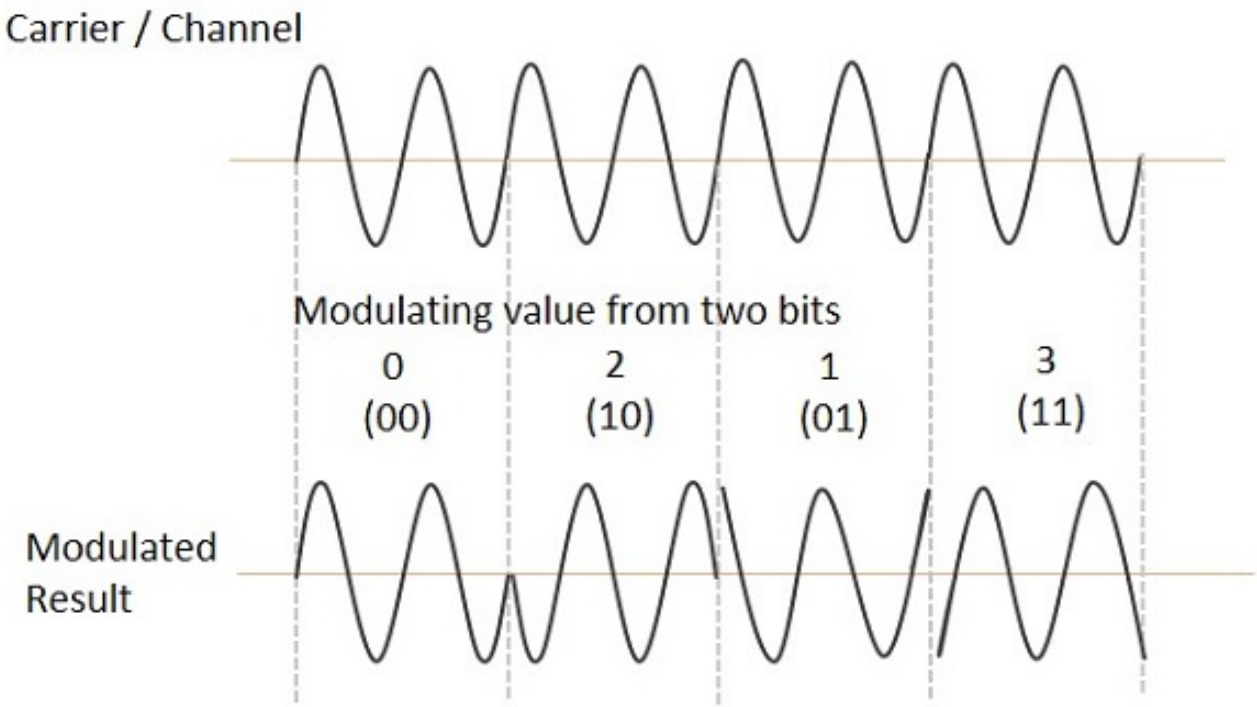
Example


Position
1
2
3
4
5
6
7
8
Bit
0
0
1
1
1
0
0
1
even bit
0
0
1
1
0
0
1
odd bit
0
0
1
1
1
1
0
0
Dibit
00
01
11
11
01
00
10
Phase
0
$\frac{\pi}{2}$
$\frac{3\pi}{2}$
$\frac{3\pi}{2}$
$\frac{\pi}{2}$
0
π

QPSK DeModulator



Waveform


Carrier / Channel
Modulating value from two bits
0
(00)
2
(10)
1
(01)
3
(11)
Modulated
Result

Non-Linear Modulation

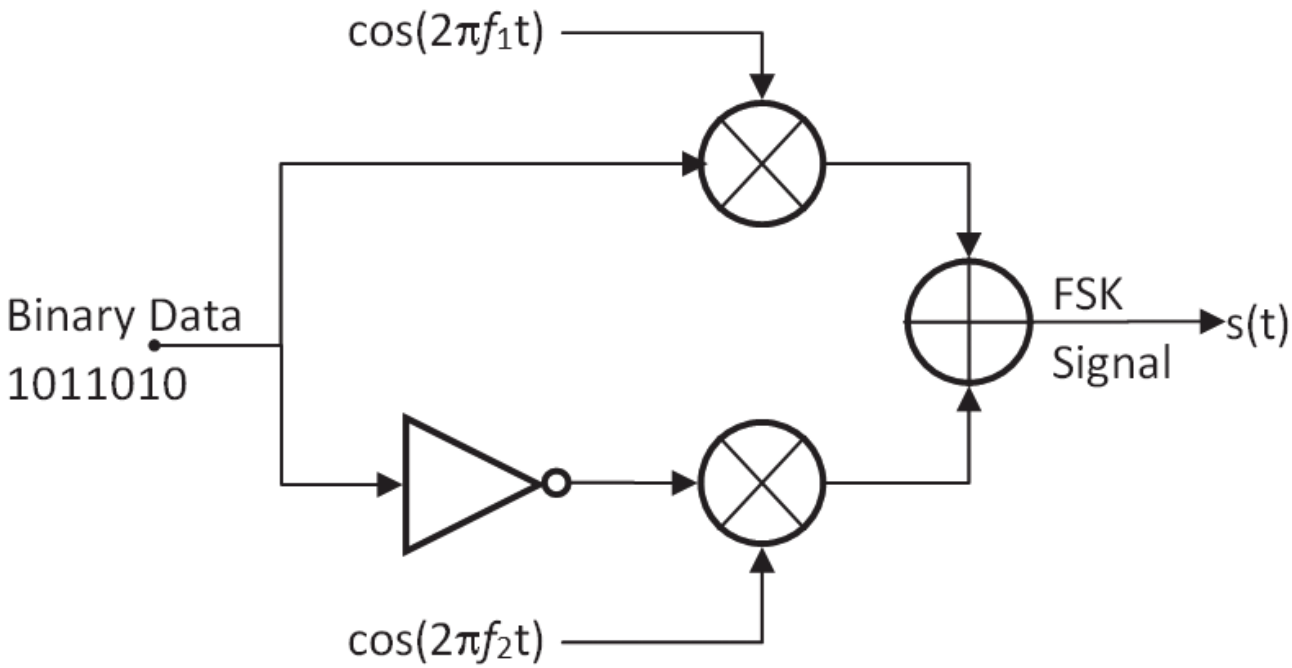
- Binary Frequency Shift Keying (BFSK)
- Minimum Shift Keying (MSK)
- Gaussian Minimum Shift Keying (GMSK)

Binary Frequency Shift Keying (BFSK)

- Binary data are represented by two signals with different frequency
- Frequency f_c of carrier is changed in response to the information
  - One frequency change encodes 0 while another frequency change encodes 1
- Carrier signal: $A_c \cos(2\pi f_c t + 0)$, assume $\phi = 0$
- The modulated signal is

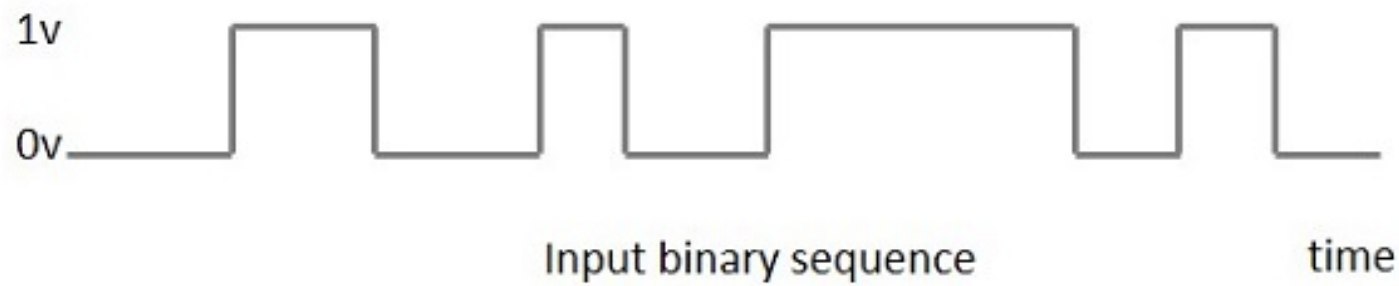
$$s(t) = \begin{cases} A_c \cos(2\pi(f_c + \Delta f)t) = A_c \cos(2\pi f_1 t) & \text{symbol "1"} \\ A_c \cos(2\pi(f_c - \Delta f)t) = A_c \cos(2\pi f_2 t) & \text{symbol "0"} \end{cases}$$

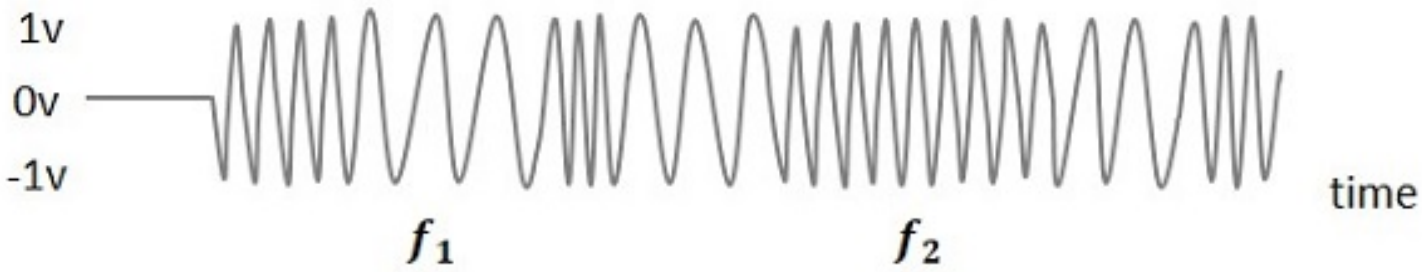
BFSK Modulation



- The two oscillators, producing a higher and a lower frequency signals, are connected to a switch
- The binary input sequence is applied to the transmitter so as to choose the frequencies according to the binary input

BFSK Modulation

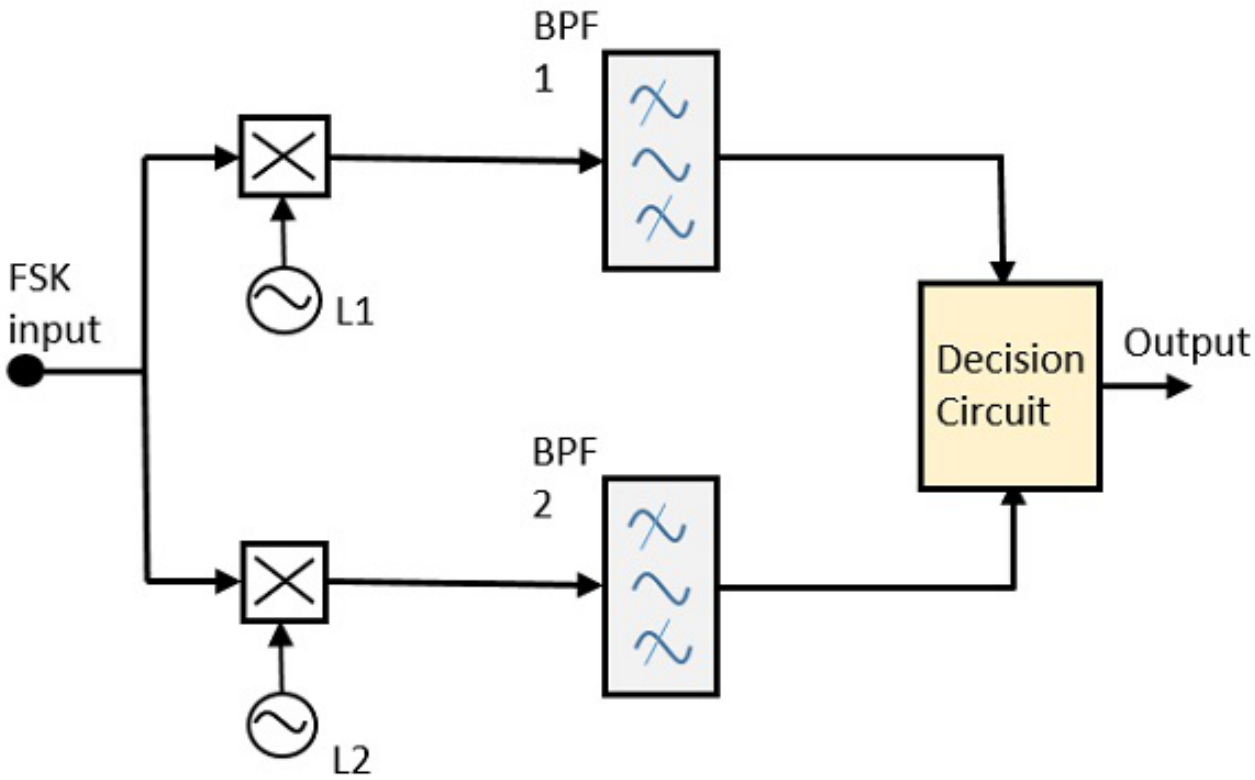




FSK Modulated output wave

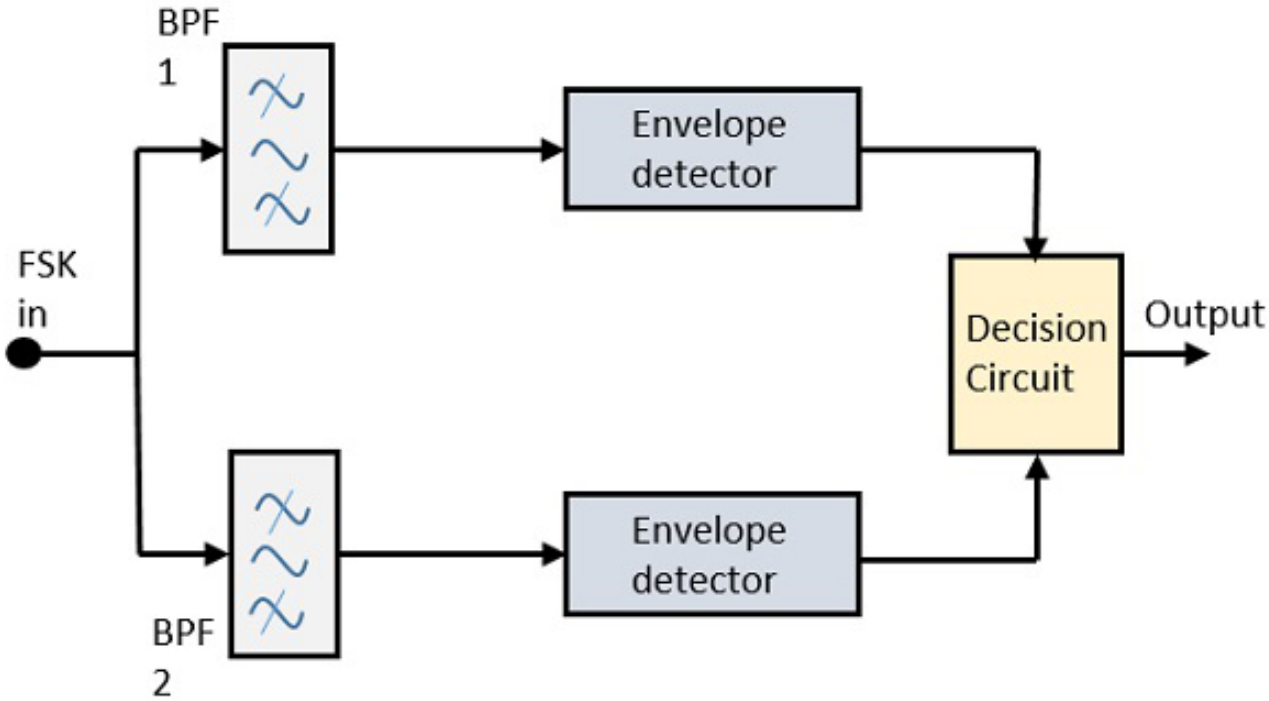
BFSK Demodulation

Synchronous FSK Detector



BFSK Demodulation

Synchronous FSK Detector



Minimum Shift Keying (MSK)

- MSK is a special type of continuous phase-frequency shift keying (CPFSK) wherein the peak frequency deviation is equal to 1/4 the bit rate R_b (or modulation index β_{MSK} is 0.5).

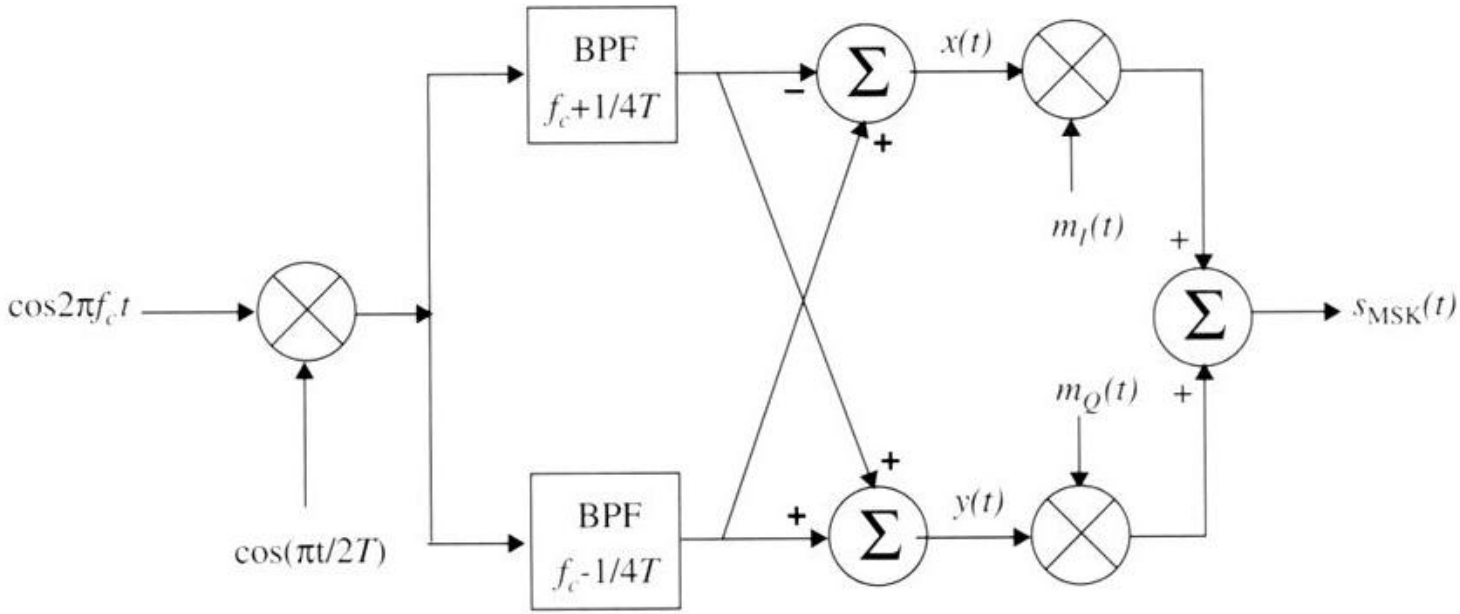
  $$\beta_{MSK} = 2\Delta f / R_b$$

- A modulation index of 0.5 corresponds to the minimum frequency spacing that allows two FSK signals to be coherently orthogonal
- Two FSK signals $s_1(t)$ and $s_2(t)$ are said to be orthogonal if

  $$\int_0^{T_b} s_1(t)s_2(t)dt = 0$$

- MSK modulation technique is very attractive for use in mobile radio communication systems due to:
  - constant envelope
  - spectral efficiency
  - good BER performance
  - self-synchronizing capability

MSK modulator



- Multiplying a carrier signal with $cos(\frac{\pi}{2T})$ produces two phase-coherent signals at $f_c + \frac{1}{4T}$ and $f_c - \frac{1}{4T}$

MSK modulator

- These two FSK signals are separated using two narrow band filters and appropriately combined to form the in-phase and quatrature carrier components $x(t)$ and $y(t)$ respectively

$$x(t) = \cos(\frac{\pi}{2T}t)\cos(2\pi f_c t)$$
$$y(t) = \sin(\frac{\pi}{2T}t)\sin(2\pi f_c t)$$

- These carriers are multiplied with the odd $(m_Q(t))$ and even $(m_I(t))$ bit streams, to produce the MSK modulated signal $S(t)$

$$\begin{aligned} S(t) &= m_I(t)x(t) + m_Q(t)y(t) \\ &= m_I(t)\cos(\frac{\pi}{2T}t)\cos(2\pi f_c t) + m_Q(t)\sin(\frac{\pi}{2T}t)\sin(2\pi f_c t) \end{aligned}$$

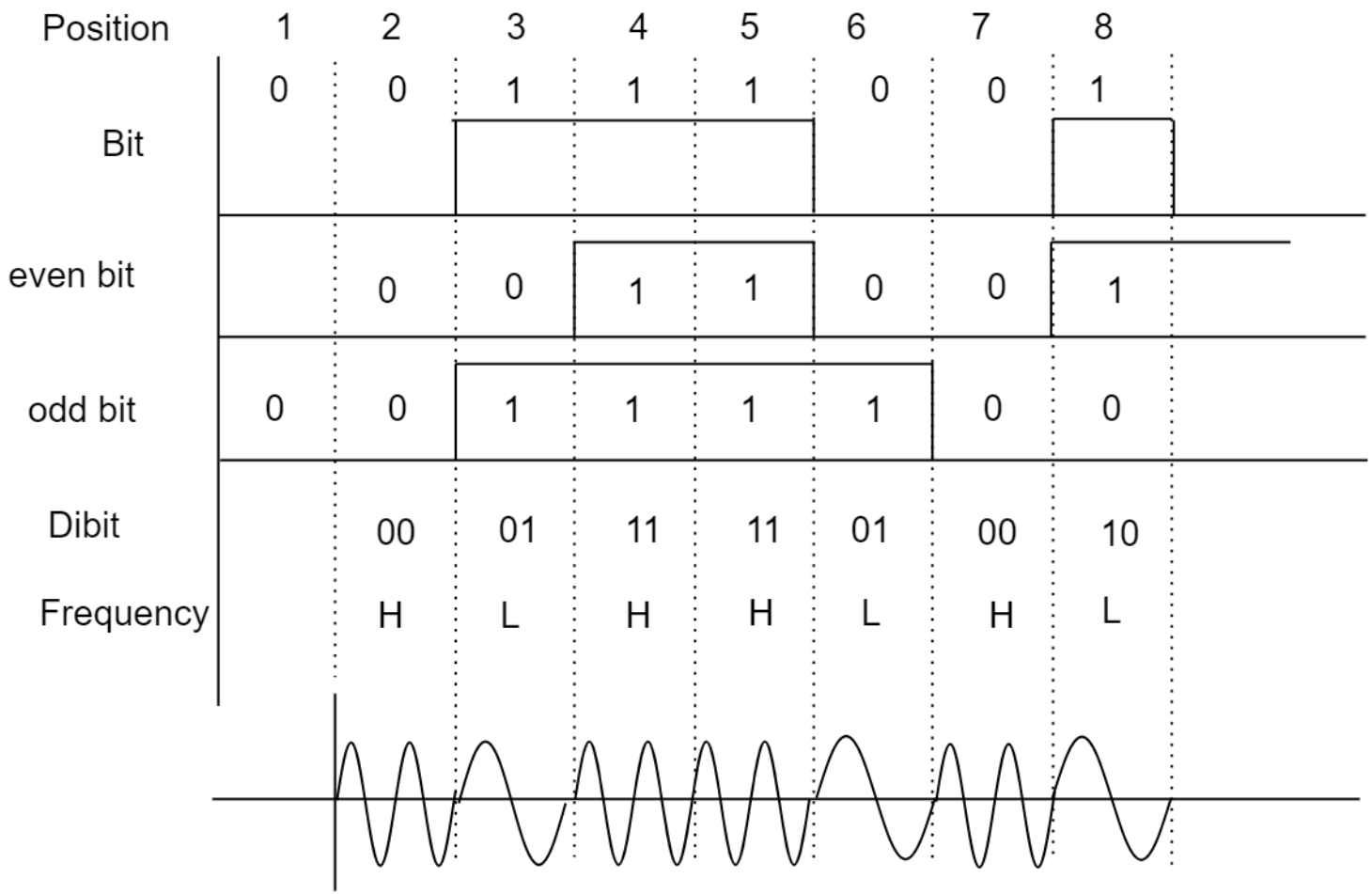
MSK Waveform

MSK generate High frequency H and Low frequency L for given bitstream based on given table

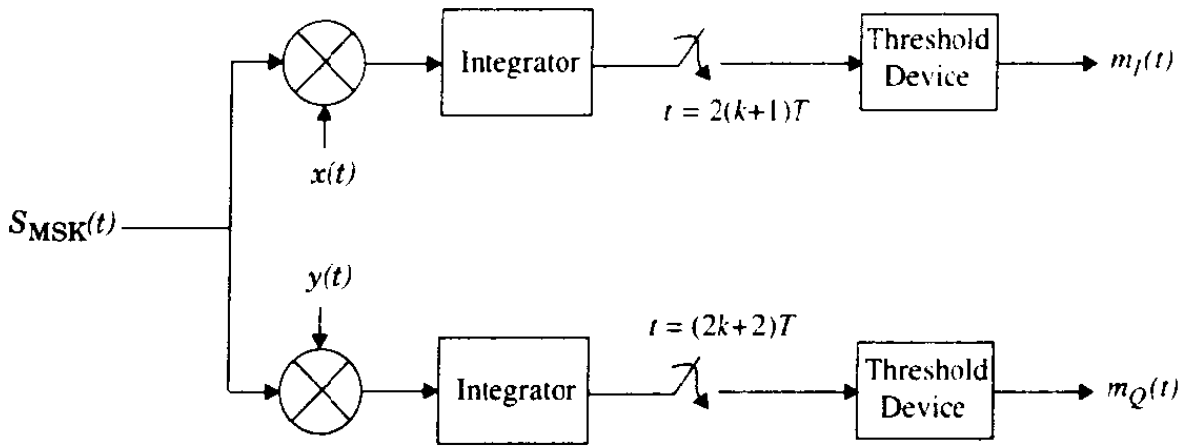
Table: Rule for MSK waveform

Even	0	1	0	1
Odd	0	0	1	1
Freq	H	L	L	H

MSK Waveform



MSK Demodulator

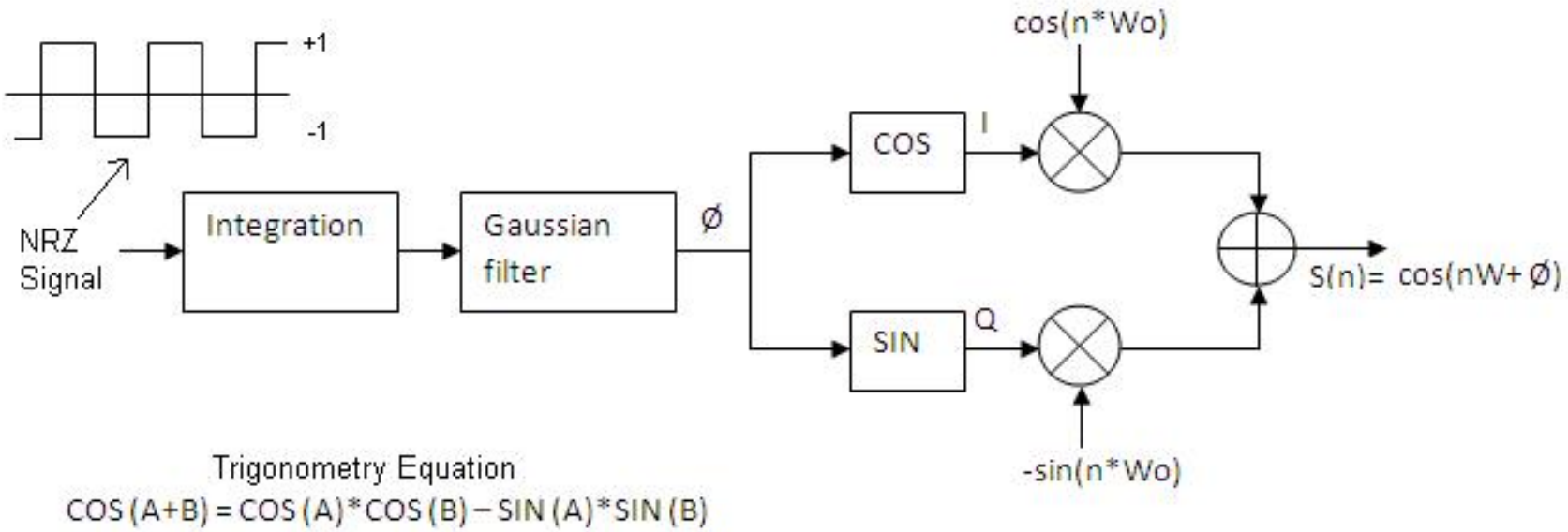


- The received signal $s(t)$ is multiplied by the respective in-phase and quadrature carriers $x(t)$ and $y(t)$
- The output of the multipliers are integrated over two bit periods
- Based on the level signal at the output of the integrator, the threshold detector decides whether the signal is a 0 or a 1
- The output data streams are $m_I(t)$ and $m_Q(t)$, which are combined to obtained the demodulation signal

Gaussian Minimum Shift Keying (GMSK)

- GMSK may be viewed as a derivative of MSK.
- In GMSK, the sidelobe levels of the spectrum are further reduced by passing the **modulating NRZ data** waveform through a **predmodulation** Gaussian pulse-shaping filter
- Baseband Gaussian pulse shaping smooths the phase trajectory of the MSK signal and hence stabilizes the instantaneous frequency variations over time.
- GMSK can be coherently detected just as an MSK signal, or noncoherently detected as simple FSK
- GMSK modulation scheme is used in GSM
  - its excellent power efficiency (due to the constant envelope) and its excellent spectral efficiency

Gaussian Minimum Shift Keying (GMSK) modulator

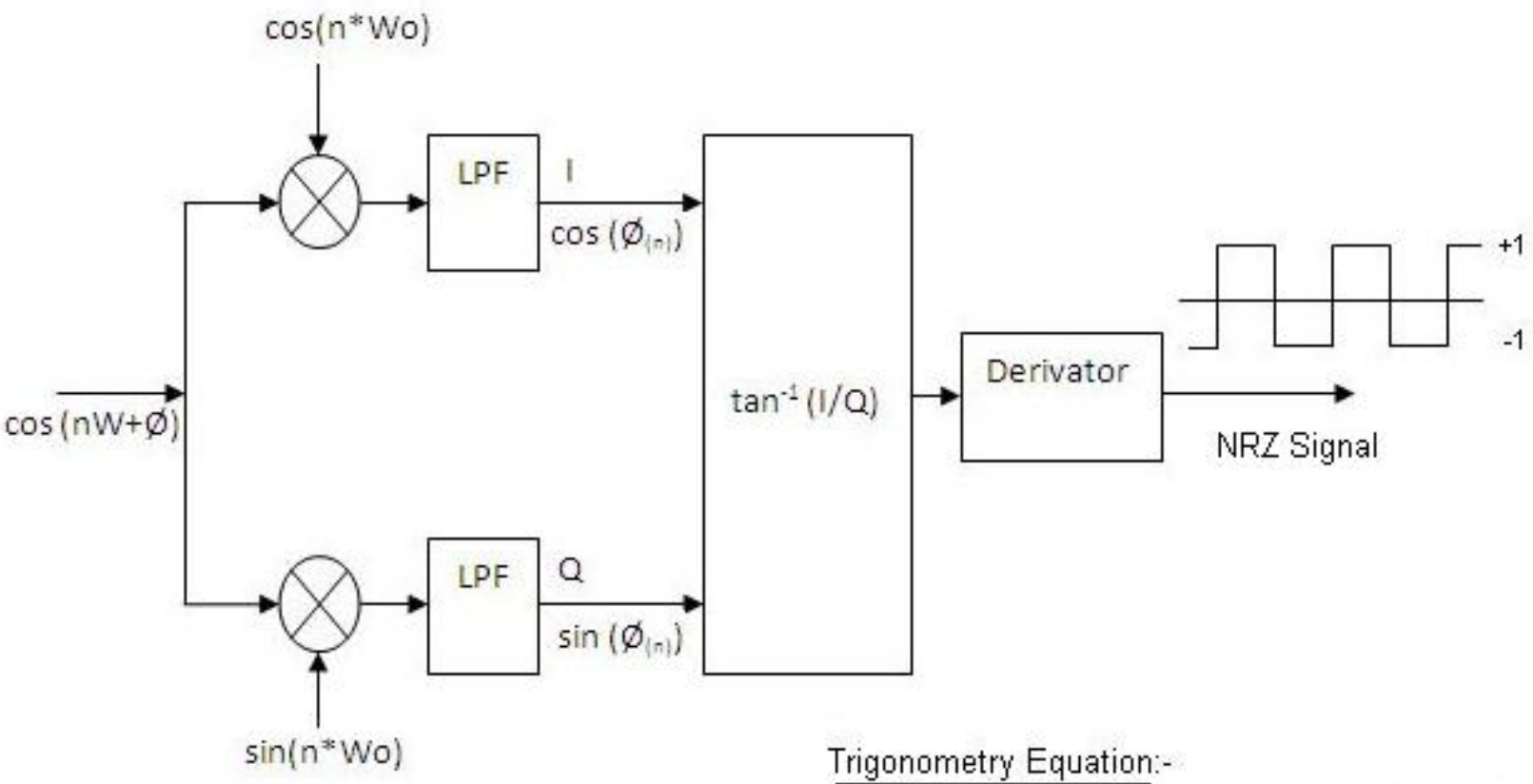


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Gaussian Minimum Shift Keying (GMSK) modulator

- Baseband Gaussian pulse shaping smooths the phase trajectory of the MSK signal and hence stabilizes the instantaneous frequency variations over time.
- GMSK can be coherently detected just as an MSK signal, or noncoherently detected as simple FSK
- GMSK modulation scheme is used in GSM
  - its excellent power efficiency (due to the constant envelope) and its excellent spectral efficiency

GMSK Demodulator



Trigonometry Equation:-

$\cos 2\theta = \cos^2\theta - \sin^2\theta \rightarrow \cos^2\theta = \cos 2\theta + \sin^2\theta$

$\sin 2\theta = 2 * \sin\theta * \cos\theta \rightarrow \sin\theta * \cos\theta = \sin 2\theta / 2$

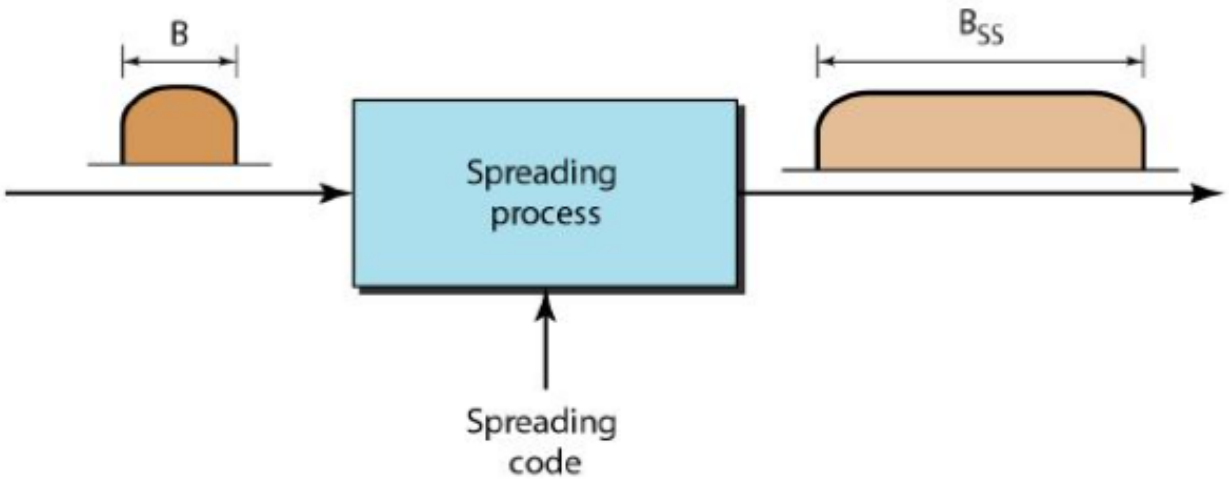
Spread Spectrum Modulation Techniques

- Due to limited bandwidth, it requires highly spectral efficiency
  - Narrow band spectrum is observed in TDMA, FDMA
- Spread spectrum modulation technique uses transmission bandwidth (B_{SS}) that is several times greater than the information signal bandwidth (B)

  $B_{SS} >> B$
  - Many users can simultaneously use the same bandwidth without significantly interfering with one another
- In a multiple-user, multiple access interference environment, spread spectrum systems become very bandwidth efficient
- Spread spectrum signals are pseudorandom and have noise-like properties

Spread Spectrum Modulation Techniques

- The spreading waveform is controlled by a pseudo-noise (PN) sequence or pseudo-noise code, which is a binary sequence that appears random but can be reproduced in a deterministic manner by intended receivers
  - PN sequence is independent of the data in the signal
  - PN sequence have
    - nearly equal number of zeros and ones
    - very low correlation between shifted versions of the sequence
    - very low cross-correlation between any two sequences



Spread Spectrum Modulation Techniques

- Spread spectrum signals are demodulated at the receiver through cross-correlation with a locally-generated version of the pseudorandom carrier
- Cross-correlation with the correct PN sequence despreads the spread spectrum signal
  - restores the modulated message in the narrow band as the original data
- Cross-correlating the signal from an undesired user results in a very small amount of wideband noise at the receiver
- Each user is assigned a unique PN code which is approximately orthogonal to the codes of other users
- All users are able to share the same spectrum
  - the receiver can separate each user based on their codes
- Spread spectrum may eliminate frequency planning
  - all cells can use the same channels
- Spread spectrum systems are not affected by multipath fading
  - the delayed versions of the transmitted PN signal will have low correlation with the original PN sequence

Processing Gain or Spreading factor

- The spread spectrum increases the bandwidth of the message signal by a factor L

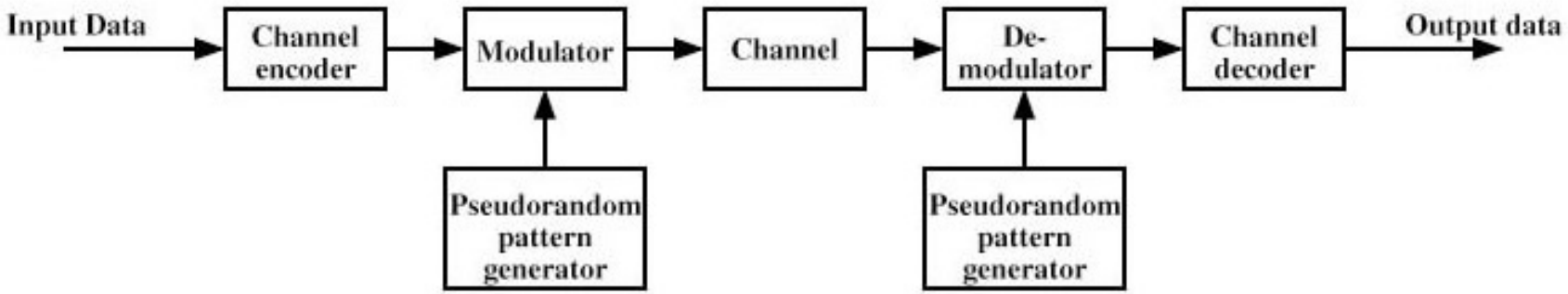
$$L = \frac{B_{SS}}{B} \quad \text{Analog System}$$

$$= \frac{T_b}{T_c} \quad \text{Digital System}$$

- where
  - B_{SS} : transmitting bandwidth
  - B : information signal bandwidth
  - T_b : bit time period
  - T_c : chip time period

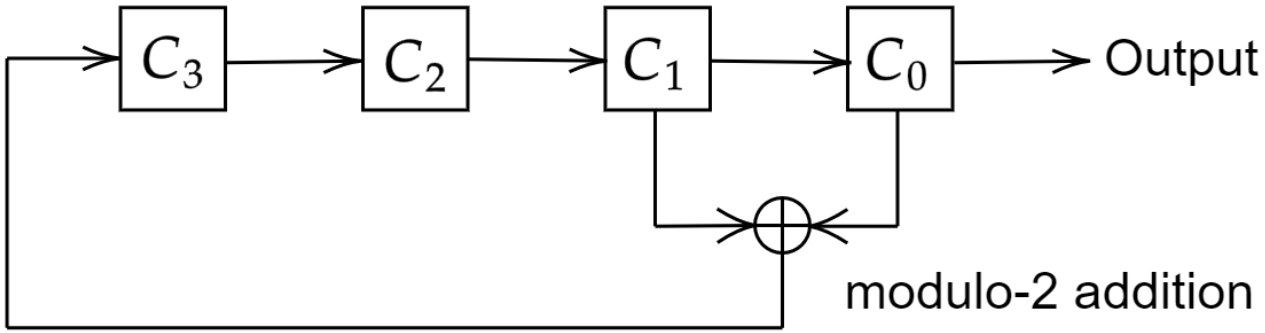
Benefits of spread spectrum

- Gain immunity from noise, multipath distortion and jamming.
- Used for hiding and encrypting signals
  - Only a recipient who knows the spreading code can recover the encoded information
- Several users can independently use the same higher bandwidth with very little interference, using code division multiple access (CDMA)



Pseudo-noise code generator

- PN code: deterministic and periodic
- Every T_C seconds, the PN source produces a new k-bit value
- Generation: Linear feedback shift register
  - all zero state not possible
  - E.g: consider a system where $k = 4$
    - The PN sequence will repeat itself with a period of $2^k - 1$
    - When $k = 4 \rightarrow 2^k - 1 = 15$



PN Code

Clock	C_3	C_2	C_1	C_0	Output	Clock	C_3	C_2	C_1	C_0	Output
1	1	0	0	0	0	9	1	0	1	0	0
2	0	1	0	0	0	10	1	1	0	1	1
3	0	0	1	0	0	11	1	1	1	0	0
4	1	0	0	1	1	12	1	1	1	1	0
5	1	1	0	0	0	13	0	1	1	1	0
6	0	1	1	0	0	14	0	0	1	1	0
7	1	0	1	1	1	15	0	0	0	1	1
8	0	1	0	1	1	16	1	0	0	0	0

- PN sequence is repeated by 15 clock
- PN code: 000100110101111

Type of Spread Spectrum

- Frequency Hopped Spread Spectrum (FH-SS): for Analog system
- Direct Sequence Spread Spectrum (DS-SS): for Digital system

Frequency Hopped Spread Spectrum (FH-SS)

- Frequency hopping involves a periodic change of carrier frequency in pseudorandom fashion
  - transmitter and receiver know the pattern and are synchronized
- A frequency hopping signal may be regarded as a sequence of modulated data bursts with time-varying, pseudorandom carrier frequencies
- The set of possible carrier frequencies is called the **hopset**

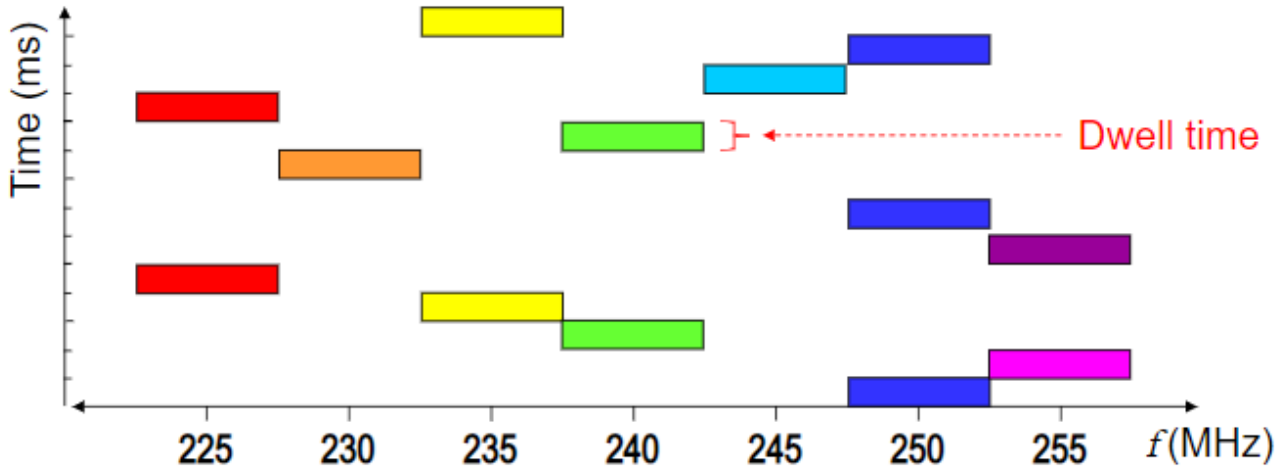


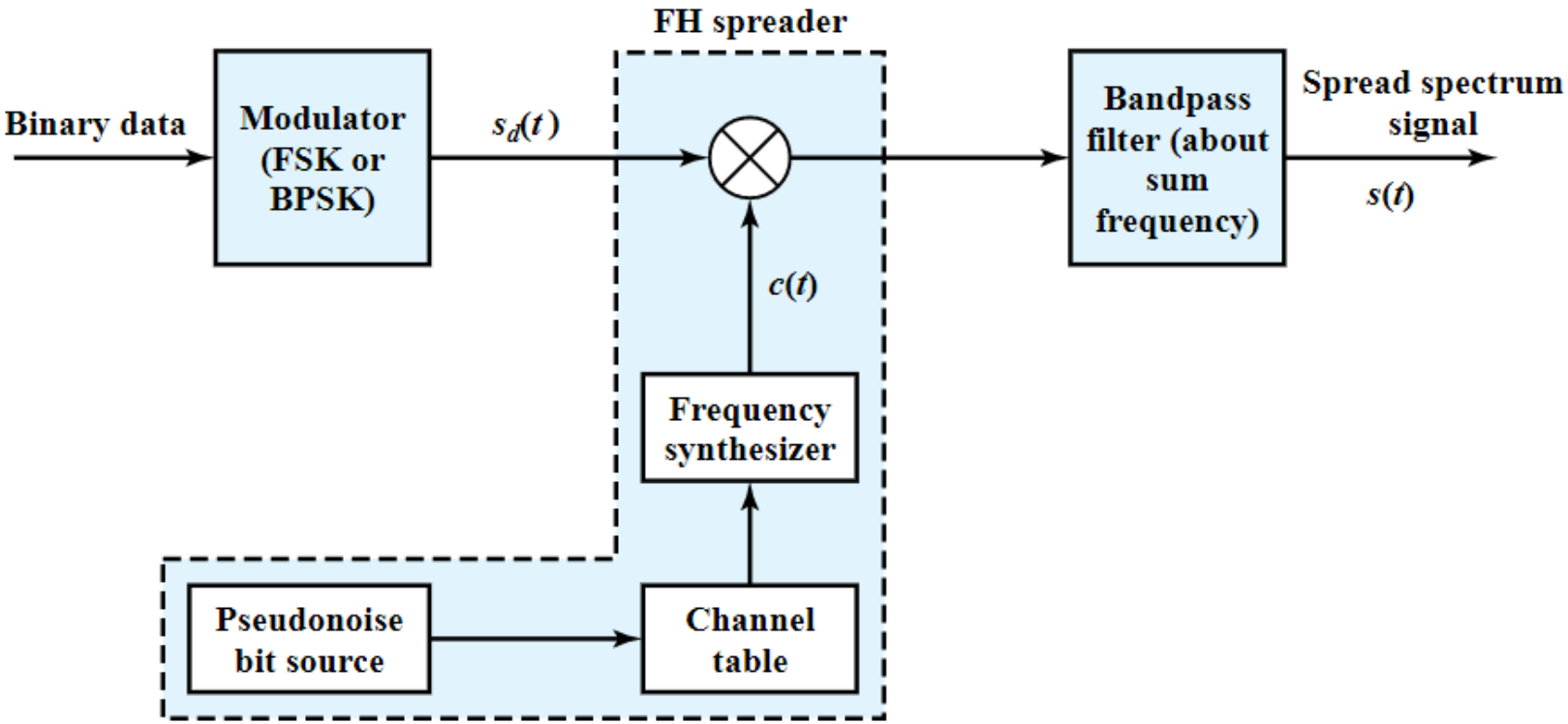
Figure:

Frequency Hopped Spread Spectrum (FH-SS)

- Hopping occurs over a frequency band that includes a number of channels
- Each channel is defined as a spectral region with a central frequency in the hopset and a bandwidth large enough to include narrowband modulation burst (usually FSK) having the corresponding carrier frequency
- **Dwell Time:** The amount of time spent on a specific frequency in an FHSS hopping sequence

  (typically $<$ 10 ms)
- **Hop duration or Hopping period (T_h)**: time duration between hops

FH-SS transmitter



FH-SS transmitter

- The input binary data sequence is applied to the M-ary FSK modulator or BPSK
- The output of FSK modulator is then applied to a mixer
  $S_d(t) = A_c \cos(2\pi(f_c + \Delta f)t)$
  - The other input to the mixer is particular frequency from frequency synthesizer
- The output of frequency synthesizer at particular instant is the frequency slot or 'hop'
  - hop is mixed with the FSK signal
- The output of the mixer is the 'sum' frequency component of FSK signal and frequency hop from synthesizer

$$\begin{aligned} S(t) &= S_d(t)c(t) \\ &= A_c \cos(2\pi(f_c + \Delta f)t)\cos(2\pi ft) \\ &= \frac{A_c}{2}[\cos(2\pi(f_c + \Delta f + f)t) + \cos(2\pi(f_c + \Delta f - f)t)] \end{aligned}$$

FH-SS transmitter

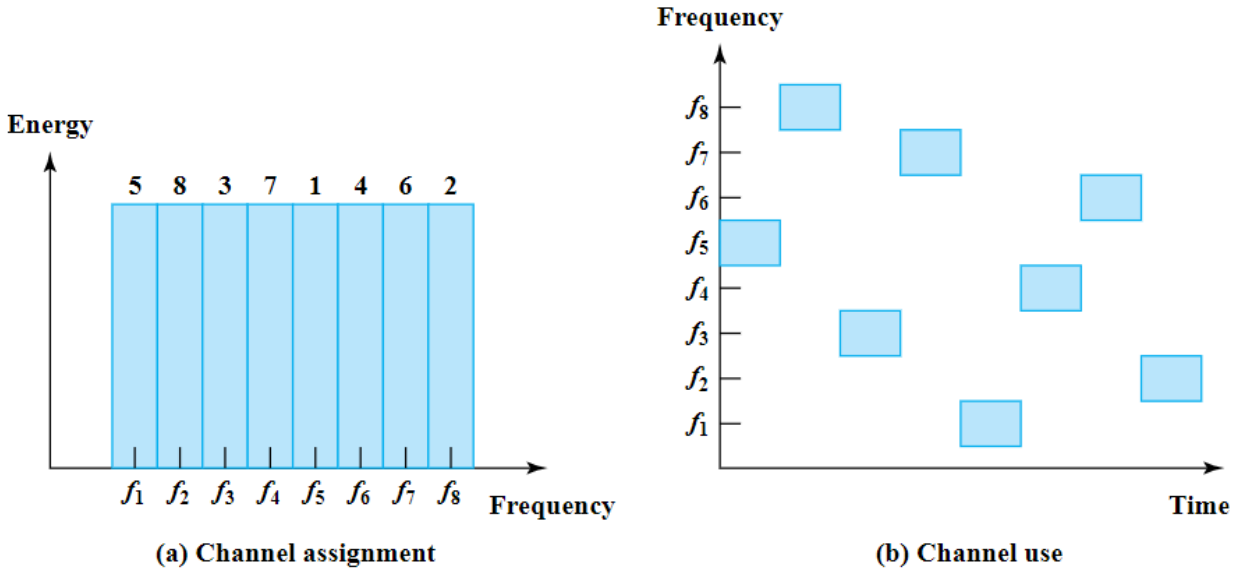
- Using a bandpass filter difference frequency can be blocked

$$S(t) = \frac{A}{2}\cos(2\pi(f_c + \Delta f + f)t)$$

- Signal is FH/MFSK signal and is transmitted over the wideband channel

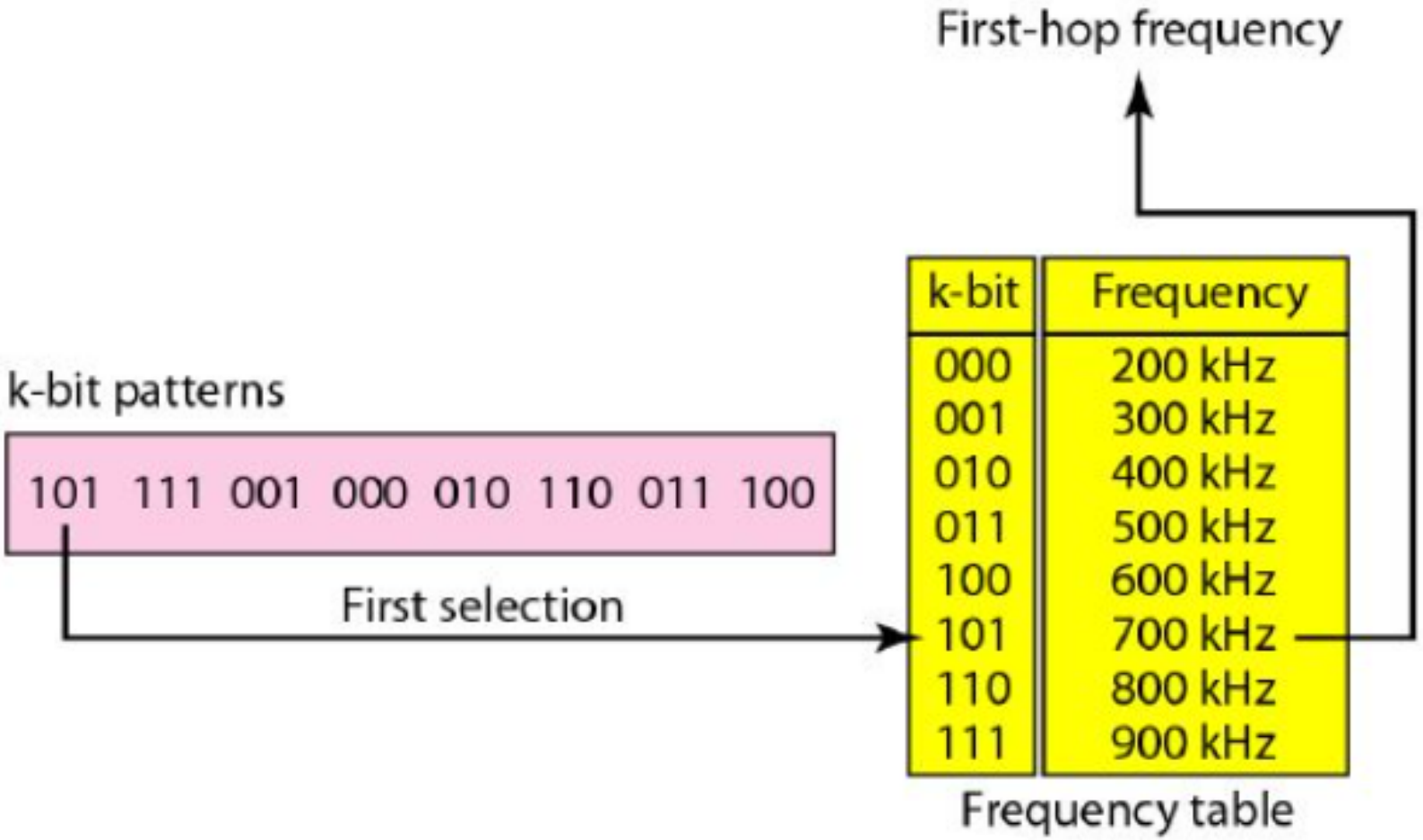
Frequency synthesizer

- A pseudo-random code generator (PN): creates a k-bit pattern for every hopping period T_h
- The frequency/ channel table uses the pattern to find the frequency to be used for hopping period and passes it to the frequency synthesizer
- The frequency synthesizer creates a carrier signal of that frequency and the source signal modulates the carrier signal

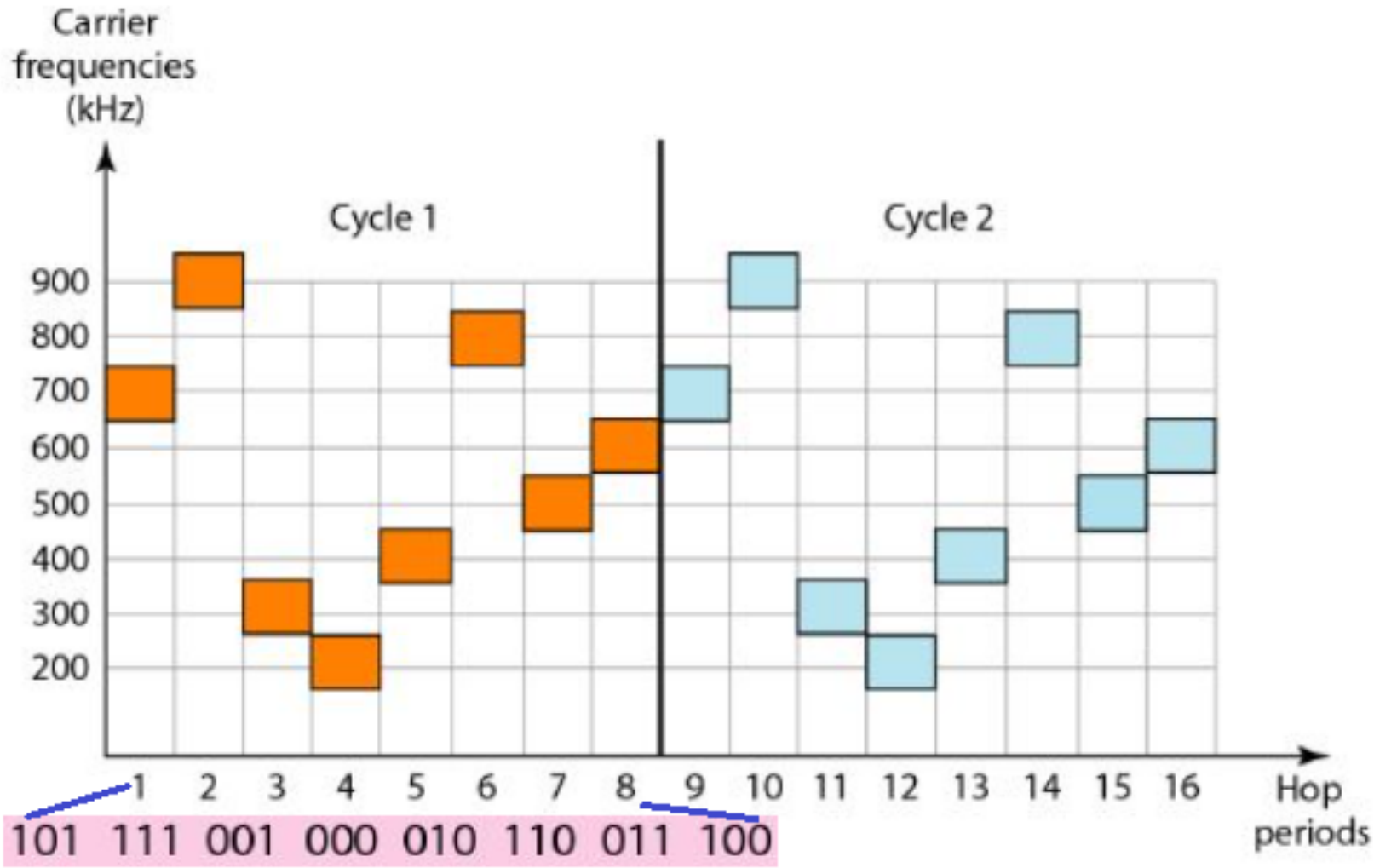


FHSS Example

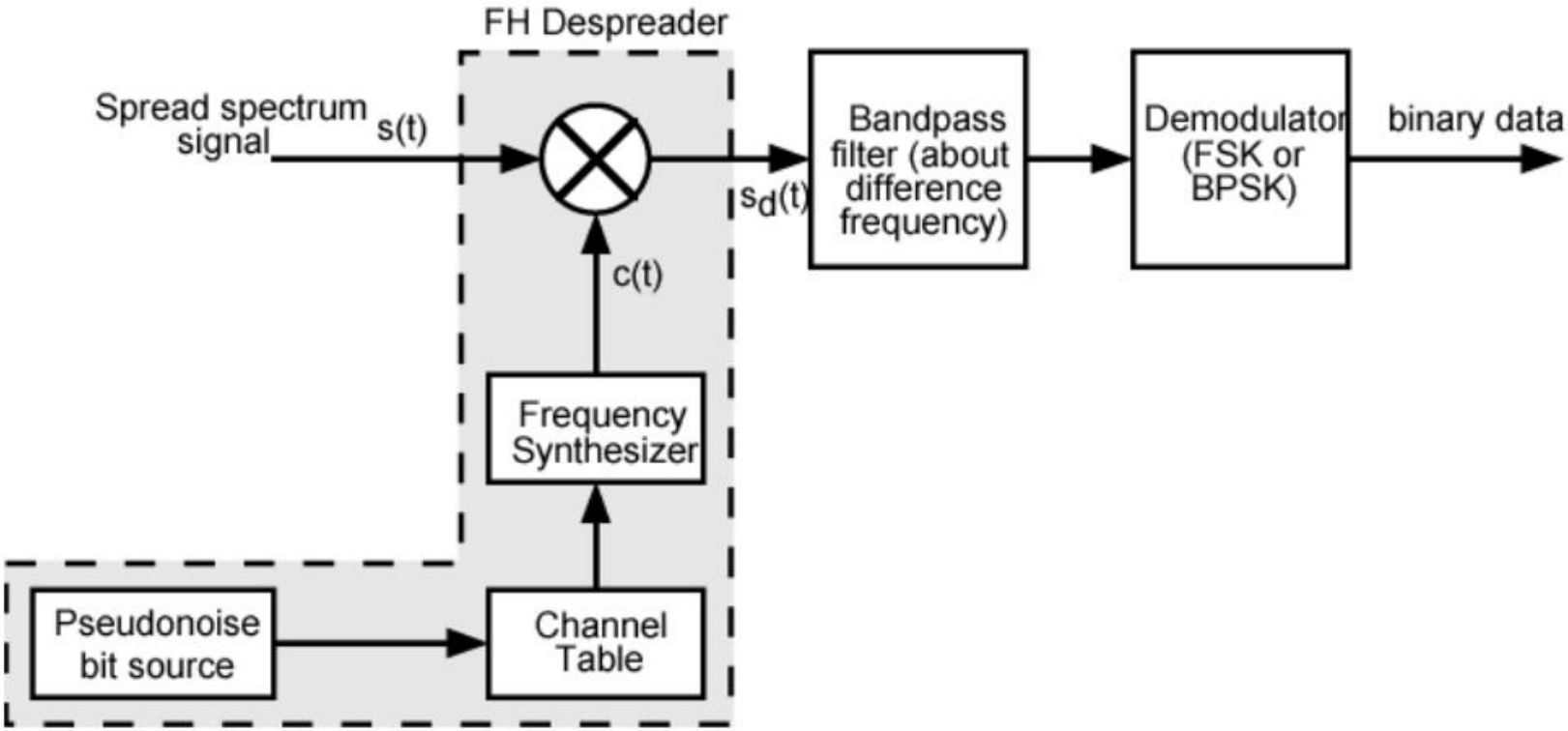
Consider M-FSK ($M = 3$)



FHSS Example



FH-SS receiver



FH-SS receiver

- At the receiver $S(t)$ is multiplied by $c(t)$

$$\begin{aligned} S_d(t) &= S(t)c(t) \\ &= \frac{A_c}{2}\cos(2\pi(f_c + \Delta f + f)t)\cos(2\pi ft) \\ &= \frac{A_c}{4}[\cos(2\pi(f_c + \Delta f + f + f)t) + \cos(2\pi(f_c + \Delta f + f - f)t)] \end{aligned}$$

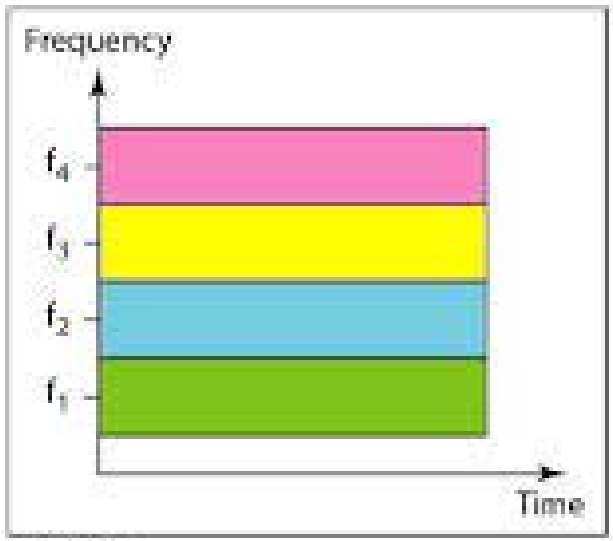
- Using a bandpass filter sum frequency can be blocked to obtained the orginal signal

$$\begin{aligned} S(t) &= \frac{A}{2}\cos(2\pi(f_c + \Delta f + f - f)t) \\ &= \frac{A}{2}\cos(2\pi(f_c + \Delta f)t) \end{aligned}$$

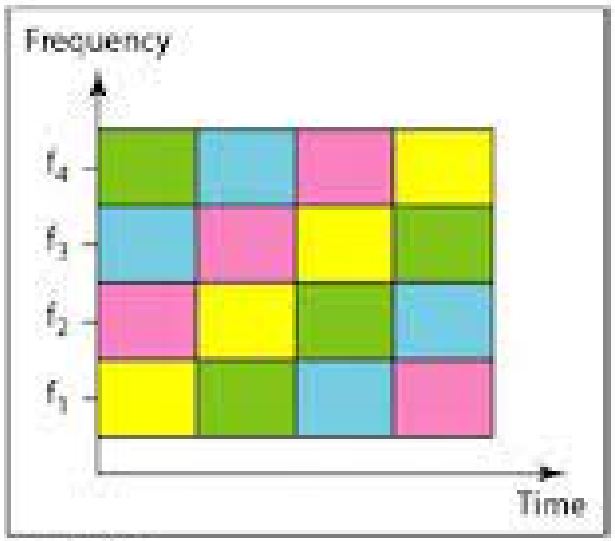
Slow and Fast FHSS

- Frequency shifted every T_c
- Duration of signal element is T_s
- Slow FHSS: $T_c \geq T_s$
- Fast FHSS: $T_c < T_s$
- Fast FHSS gives improved performance in noise or jamming

FDM vs FHSS



a. FDM



b. FHSS

- Let number of hoping frequencies be M
  - M channel multiplex into one: using B_{SS}
- In FDM, each station uses $1/M$ of bandwidth and allocation is fixed
- In FHSS, each station uses $1/M$ of bandwidth but allocation changes at every hop

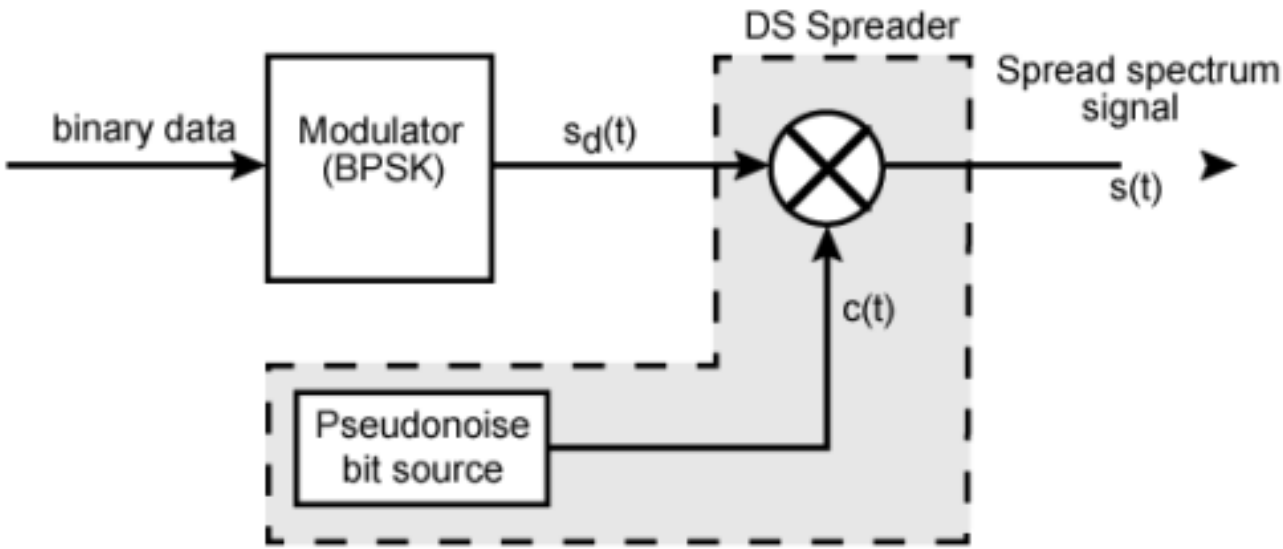
Direct Sequence Spread Spectrum (DS-SS)

- Technique in which the data (message) signal is multiplied with a pseudo-random spreading sequence that has a much higher chip rate R_c than the original data rate R_b

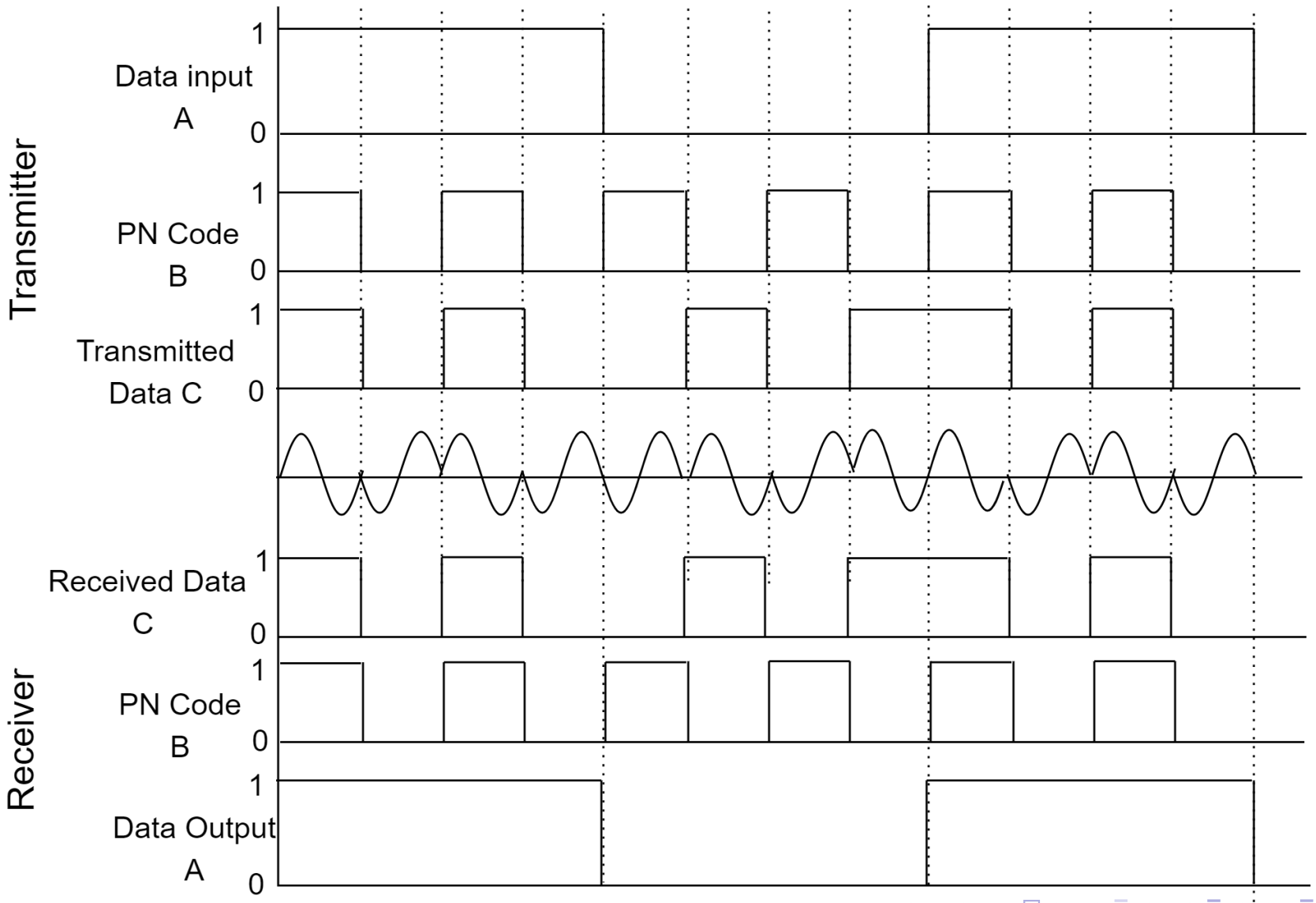
  $$R_c >> R_b$$

- The spread spectrum signal is PSK modulated.
- Each bit is assigned a code of n bits, called chips, where the chip rate is n times that of the data bit.

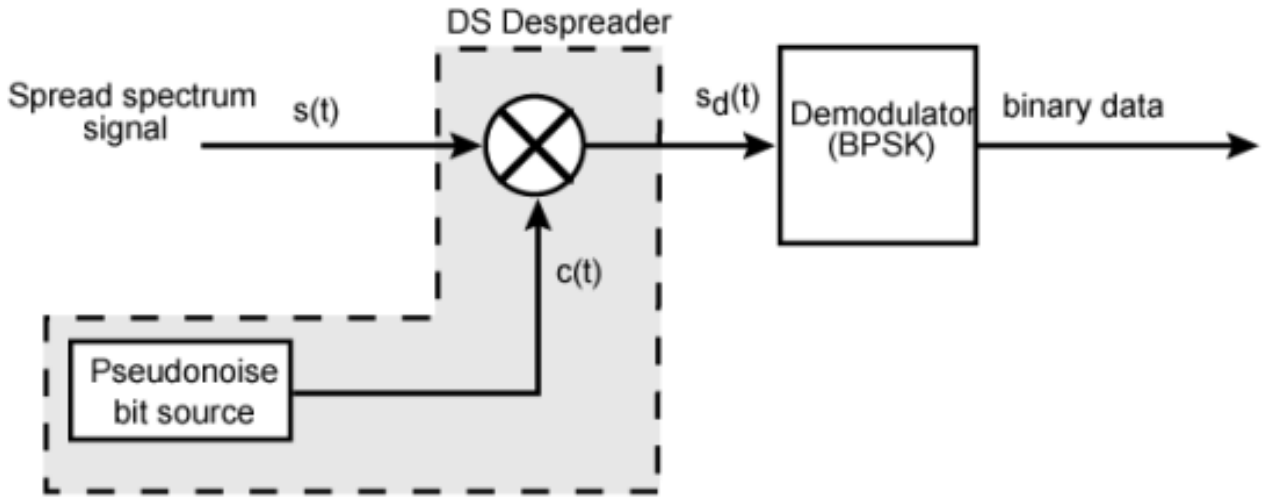
DS-SS Transmitter



DS-SS Waveform



DS-SS Receiver



Orthogonal Frequency Division Multiplexing (OFDM)

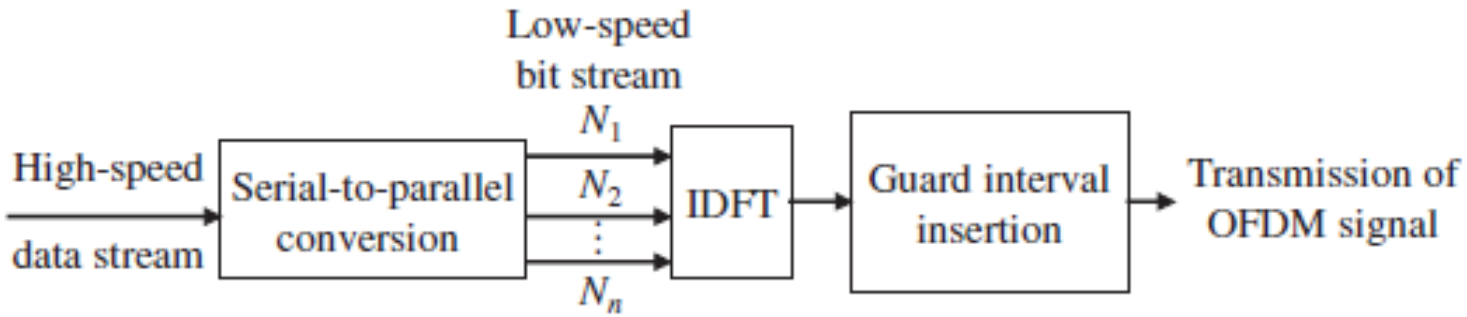
- Modulation scheme that is especially suited for high-data-rate transmission in delay-dispersive (frequency selective) environments
- Converts a high-rate data stream into a number of low-rate streams that are transmitted over parallel, narrowband channels that can be easily equalized
- It is a digital multi-carrier modulation scheme that extends the **concept of single subcarrier modulation** by using multiple subcarriers within the same single channel
- OFDM makes use of a large number of closely spaced orthogonal subcarriers that are transmitted in parallel
- Each subcarrier is modulated with a conventional digital modulation scheme (such as QPSK, 16QAM, etc.) at low symbol rate
- The combination of many subcarriers enables data rates similar to conventional single-carrier modulation schemes within equivalent bandwidths.

OFDM Transmitter

- The orthogonality condition of the two signals in OFDM can be given by

$$\int_F s_i(f,t)s_j^*(f,t)dt = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$$

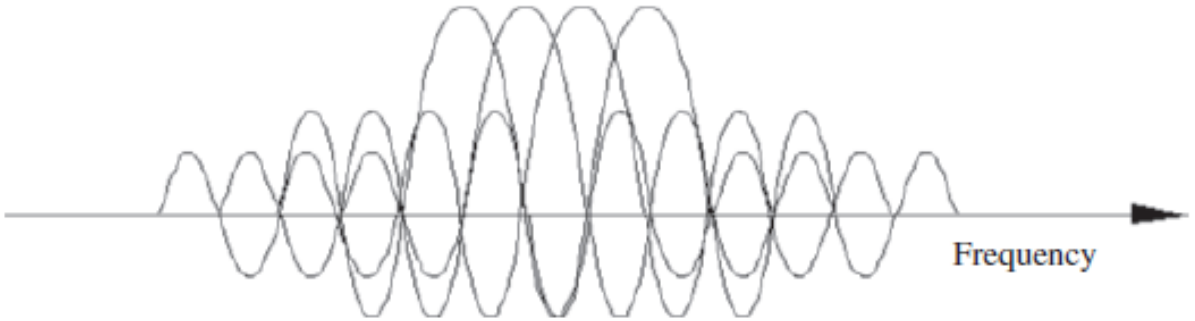
- OFDM is based on the well-known technique of Frequency Division Multiplexing (FDM)
- In FDM different streams of information are mapped onto separate parallel frequency channels
- Each FDM channel is separated from the others by a frequency guard band to reduce interference between adjacent channels



OFDM Waveform

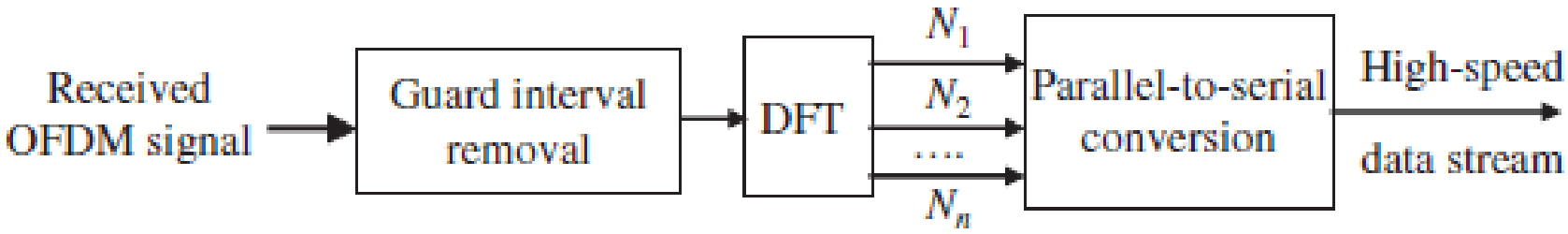


(a) Single OFDM subchannel



(b) An OFDM signal with multiple subchannels

OFDM Receiver



Speech Coding

- In mobile communication systems, bandwidth is a precious commodity, and service providers are continuously met with the challenge of accommodating more users within a limited allocated bandwidth
- The **lower** the bit rate at which the coder can deliver toll quality speech, the **more** speech channels can be compressed within a given bandwidth
- The **performance** of speech coders determines the quality of the recovered speech and the **capacity** of the system
- The goal of all speech coding systems:
  - to transmit speech with the highest possible quality using the least possible channel capacity
  - process to convert speech signal with higher bit rate to lower bit rate, i.e., **speech compression process**
- It is a procedure to represent a digitized speech signal using as few bits as possible while maintaining the speech quality
  - Speech codec in mobile phone technology

Speech Coding



Characteristics of Speech Signals

Speech waveforms have a number of useful properties that can be exploited when designing efficient coders

- Non uniform probability distribution of speech amplitude
- Non zero autocorrelation between successive speech samples
- Non flat nature of the speech spectra
- Existence of voiced and unvoiced segments in speech
- Quasi-periodicity of voiced speech signals

1. **Probability Density Function (pdf)**
   - Very high probability of near-zero amplitudes
   - Significant probability of very high amplitudes
   - Monotonically decreasing function of amplitudes between these extremes amplitudes

Characteristics of Speech Signals

2. **Autocorrelation Function (ACF)**
   - There exists much correlation between adjacent samples of a segment of speech
     - allow easily predicting
     - All differential and predictive coding schemes are based on ACF
   - ACF gives a quantitative measure of the closeness between samples.
   - Typical signals have an adjacent sample correlation, C(1) , as high as 0.85 to 0.9.
3. **Power Spectral Density Function (PSD)**
   - PSD is non flat
     - High frequency components contribute very little to the total speech energy
   - Can be used to obtain significant compression in frequency domain
   - Coding speech separately in different frequency bands can lead to significant coding gain
     - Though high frequency is insignificant in energy, they are very **important** carriers of speech information, and hence need to be adequately represented

Frequency Domain Coding of Speech

- Speech signal is divided into a set of **frequency components** which are quantized and encoded separately.
- Different frequency bands can be **preferentially encoded** according to some **perceptual** criteria for each band.
  - **Advantage:** The number of bits used to encode each frequency component can be **dynamically varied** and shared among the different bands.
- The quantization noise can be contained within bands and prevented from creating harmonic distortions outside the band
- E.g: Sub-band coding (SBC), Adaptive transfer coding (ATC)

Sub-band coding (SBC)

- Speech is typically divided into four or eight sub-bands by a bank of filters

Sub-band	Frequency Range
1	$200 - 700$ Hz
2	$700 - 1310$ Hz
3	$1310 - 2020$ Hz
4	$2020 - 3200$ Hz

- Each subband is sampled at a bandpass Nyquist rate and encoded with different accuracy in accordance to a perceptual criteria
- Useful for **lower bit rates** in the range from 9.6 Kbps to 32 Kbps
  - Especially when bit rate below 16kbps
    - The CD-900 cellular telephone system uses sub-band coding speech compression.

Adaptive Transform Coding

- Make the transformations of windowed input segments of the speech waveform
  - Each segment is represented by a set of transform coefficients, which are separately quantized and transmitted
  - Discrete cosine transform (DCT) is usually used to implement the transform
- Successfully used to encode speech at bit rates in the range 9.6 Kbps to 20 Kbps.

Classification: Based on compression

1. Waveform Coders
2. Vocoders

Waveform Coders

- reproduce the time waveform of the speech signal as closely as possible.
- Source independent
- Code equally well a variety of signals.
- Robust for a wide range of speech characteristics and for noisy environments.
- With minimal complexity
- Achieves only moderate economy in transmission bit rate.
- Examples:
  - Pulse code modulation (PCM)
  - Adaptive differential pulse code modulation (ADPCM)
  - Delta modulation (DM)

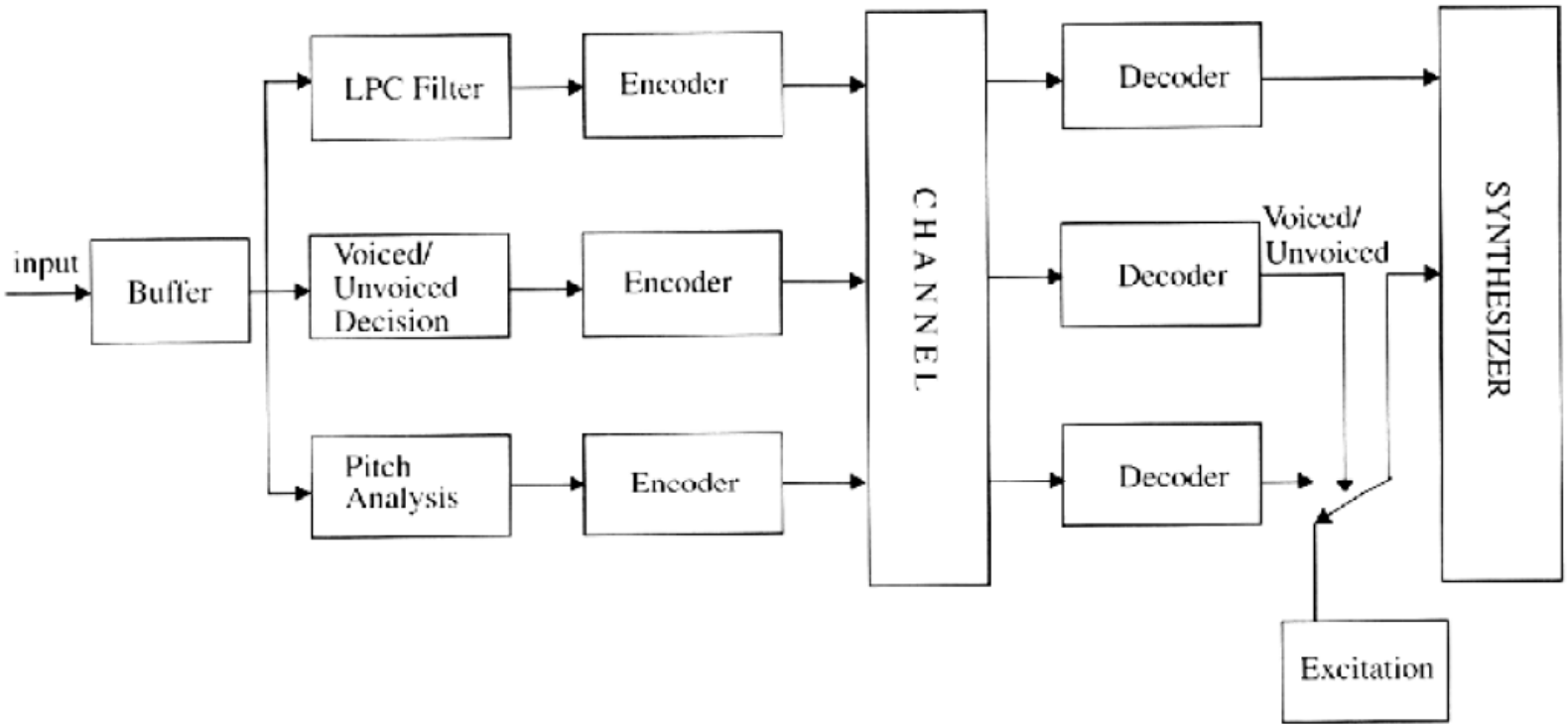
Vocoders

- Coding systems that analyze the voice signal at the transmitter
  - transmit parameters derived from the analysis, and then synthesize the voice at the receiver using those parameters
- Much more complex than the waveform coders
- Achieve **very high** economy in transmission bit rate.
- Less robust
- Types:
  - Linear predictive coder (LPC)
  - Channel vocoder
  - Formant vocoder
  - Cepstrum vocoder
  - Voice excited vocoder

Linear predictive coder (LPC)

- Belong to the time domain class of vocoders.
- Attempts to extract the significant features of speech from the time waveform.
- Computationally intensive, but most popular among the class of low bit rate vocoders
  - Transmit good quality voice at 4.8 kbps and poorer quality voice at even lower rates
- The prediction principles are similar to those in ADPCM, but LPC transmits only selected characteristics of the error signal, includes:
  - Gain factor
  - Pitch information
  - Voiced/unvoiced decision information.
- Models the vocal tract as an all pole linear filter
  - **excitation** to the filter is either a pulse at the pitch frequency or random white noise depending on whether the speech segment is **voiced or unvoiced**
  - The coefficients of the all pole filter are obtained in the time domain using linear prediction techniques

LPC


input
Buffer
LPC Filter
Voiced/
Unvoiced
Decision
Pitch
Analysis
Encoder
Encoder
Encoder
CHANNEL
Decoder
Decoder
Decoder
Voiced/
Unvoiced
SYNTHESIZER
Excitation

Multi-pulse Excited LPC

- No matter how well the pulse is positioned, excitation by a single pulse per pitch period produces audible distortion.
- Using more than one pulse, typically eight per period, and adjusting the individual pulse positions and amplitudes sequentially to minimize a spectrally weighted mean square error.
  - Can results in better speech quality, because the prediction residual is better approximated by several pulses per pitch period
- The multi-pulse algorithm does not require pitch detection.

Code-Excited LPC

- The coder and decoder have a predetermined code book of stochastic (zero-mean white Gaussian) excitation signals.
- For each speech signal the transmitter searches through its code book of stochastic signals for the one that gives the best perceptual match to the sound when used as an excitation to the LPC filter.
- The index of the code book where the best match was found is then transmitted.
- The receiver uses this index to pick the correct excitation signal for its synthesizer filter.
- Extremely complex, but can provide high quality even when the excitation is coded at only 0.25 bits per sample.
- CDMA digital cellular standard (IS-95): variable rate CELP codec at 1.2 to 14.4 kbps, and QCELP13 at 13.4 kbps

Residual Excited LPC

- In this class of LPC coders, after estimating the model parameters (LP coefficients or related parameters) and excitation parameters (voiced/unvoiced decision, pitch, gain) from a speech frame, the speech is synthesized at the transmitter and subtracted from the original speech signal to from a residual signal.
- The residual signal is quantized, coded, and transmitted to the receiver along with the LPC model parameters.
- At the receiver the residual error signal is added to the signal generated using the model parameters to synthesize an approximation of the original speech signal.
- The quality of the synthesized speech is improved due to the addition of the residual error

GSM Codec

- Regular pulse excited long term prediction (RPE-LTP) codec
- speech coder used in the pan-European digital cellular standard GSM, was chosen after conducting exhaustive subjective tests on various competing codecs
- Net bit rate: 13 kbps.
- RPE-LTP is a combination of two proposed codec:
  1. baseband RELP codec
     - Advantage: provides good quality speech at low complexity.
     - Drawback: affected by channel errors.
  2. multi-pulse excited long-term prediction (MPE-LTP) codec
     - Advantage: produces excellent speech quality, not much affected by bit errors in the channel.
     - Drawback: high complexity.
- By modifying the RELP codec to incorporate certain features of the MPE-LTP codec, the net bit rate was reduced from 14.77 kbps to 13.0 kbps without loss of quality
- The GSM codec is relatively complex and power hungry.